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Determinants of Occupational Mobility in England and Wales, 1861-1911: A Census-Linking Approach

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Abstract

As fears over automation and globalisation become centre stage in economic and foreign policy, understanding how workers respond to periods of significant economic upheaval is crucial to our ability to make informed, impactful policy. In the past twenty years, a significant body of work has emerged to disentangle the local labour market impacts of technological innovation and trade competition. However, modern data offer little insight into the long- and medium-term effects of economic change, especially as they relate to occupational mobility and worker welfare. This dissertation examines these labour market dynamics during a period of rapid economic change using a novel dataset of 4.4 million linked census records of adult men in England and Wales from 1861 to 1911. In examining individual mobility across 20-year samples, I find that age and region play a significant role in determining access to opportunity. I then examine mobility twenty years after an adverse trade shock to Britain's largest industry—cotton textile manufacturing. I find that workers employed in cotton prior to the shock exhibit long-term diminished mobility compared to non-cotton workers. These results suggest that the factors underpinning mobility merit further examination, especially for workers exposed to adverse shocks. In examining individual outcomes across three 20-year linked samples, this research disentangles drivers of occupational upgrading from 1861 to 1911, both across cohorts and following an adverse trade shock localised to one industry.

1 Introduction

As AI and trade reshape labor markets, understanding worker adjustment across the life course is critical for designing effective policy. Although the aggregate, long-run effects of job decline may be negligible, the consequences for workers over their lifetimes can be significant. Despite the importance of individual outcomes during periods of economic upheaval, little is known about how economic change affects worker mobility across the life course. Understanding these life-cycle effects is crucial to creating policies that promote worker welfare and adapt dynamically to evolving market demands.

Historical data offer a compelling insight into this issue, as these data allow researchers to examine the individual impact of labour market change across the life cycle. From the late eighteenth to the nineteenth century, Britain experienced unprecedented structural change. This period, which covers both the First and Second Industrial Revolutions, witnessed widespread mechanisation, the expansion of textile factories, and the rise of urban labour markets.[1] This transformation came with human costs, as technological unemployment led to profound employment losses among certain workers.[2] Similar analyses of job displacement across varying sectors illustrate that employment shocks often imposed severe short- and medium-term costs on workers.[3]

This project creates a novel database of over four million linked census records of adult men in England and Wales from 1861 to 1911. During this period, transaction costs fell rapidly while new forms of labour management emerged.[4] While significant work has examined how technological change affects employment, there have been few large-scale analyses of life course trends in occupational mobility, especially after an adverse shock. Understanding these patterns is especially important given parallels between trade and job decline in American manufacturing and British cotton textiles.[5]

By focusing on individual-level experiences, this project sheds light on mobility through-

1. Robert Allen, *The British industrial revolution in global perspective*, New approaches to economic and social history (Cambridge: Cambridge University Press, 2009), pp. 1-22, ISBN: 978-0-511-81668-0 978-0-521-68785-0 978-0-521-86827-3, doi:10.1017/CBO9780511816680; Robert Allen, "The Industrial Revolution in Miniature: The Spinning Jenny in Britain, France, and India," *The Journal of Economic History* 69, no. 4 (December 2009): pp. 901-27, ISSN: 0022-0507, 1471-6372, doi:10.1017/S0022050709001326Eric J. Hobsbawm, *The age of revolution: 1789-1848*, First Vintage Books edition (New York: Vintage Books, 1996), pp. 27-52, ISBN: 978-0-679-77253-8.

2. Jane Humphries and Benjamin Schneider, "Spinning the industrial revolution," *The Economic History Review* 72, no. 1 (February 2019): pp. 126-55, ISSN: 0013-0117, 1468-0289, doi:10.1111/ehr.12693

3. Louis S. Jacobson, Robert J. LaLonde, and Daniel G. Sullivan, "Earnings Losses of Displaced Workers," Publisher: American Economic Association, *The American Economic Review* 83, no. 4 (1993): p. 697, ISSN: 00028282; David Autor, David Dorn, and Gordon Hanson, "On the Persistence of the China Shock," Publisher: Brookings Institution Press, *Brookings Papers on Economic Activity*, 2021, p. 384, ISSN: 00072303, 15334465

4. Maxine Berg, *The Age of Manufactures, 1700-1820: Industry, Innovation and Work in Britain*, 2nd ed (Hoboken: Taylor / Francis, 2005), 155-193, ISBN: 978-0-415-06934-2; Yrjö Kaukiainen, "Shrinking the world: Improvements in the speed of information transmission, c. 1820-1870," *European Review of Economic History* 5, no. 1 (April 2001): pp. 1-28, ISSN: 1361-4916, 1474-0044, doi:10.1017/S1361491601000016

5. William Lazonick, "Industrial Organization and Technological Change: The Decline of the British Cotton Industry," *The Business History Review* 57, no. 2 (1983): 195–236, ISSN: 00076805; David Autor, David Dorn, and Gordon Hanson, "The China Syndrome: Local Labor Market Effects of Import Competition in the United States," *American Economic Review* 103, no. 6 (October 2013): 2121–2168, ISSN: 0002-8282, doi:10.1257/aer.103.6.2121; David Autor, David Dorn, and Gordon Hanson, "When Work Disappears: Manufacturing Decline and the Falling Marriage-Market Value of Young Men," *National Bureau of Economic Research Working Paper Series* No. 23173 (2017), doi:10.3386/w23173

out the life cycle. Linking census records across decades offers novel insight into how workers navigated industrial change, including recovery from localised trade shocks and the long-term consequences of economic change.

1.1 Motivation

A central question in economic history is how people adapt to economic change. Existing approaches to answering this question have an important limitation: modern data allow researchers to track workers only for a decade or so, typically through survey panels, employer-employee links, or administrative records. These methods provide insight into short-run outcomes, but do not capture how workers adjust over the life cycle.

The digitisation and standardisation of historical census records have opened a window into these questions. Linked census data allow researchers to follow millions of individuals across decades, observing long-run occupational mobility, geographic relocation, and family outcomes at a scale and level of precision previously impossible.[6] By embedding short-run displacements in a life-cycle frame, linked census analysis can reveal whether initial losses represent temporary disruptions or lasting scars.[7]

This dissertation constructs a sample of over 4.4 million linked adult male census records in England and Wales, 1861 to 1911, to assess how trade and economic change impact occupational mobility. Throughout the nineteenth and early twentieth centuries, Britain underwent a period of dramatic transformation, marked by sweeping technological, social,

6. Joseph P. Ferrie, "A New Sample of Males Linked from the Public Use Microdata Sample of the 1850 U.S. Federal Census of Population to the 1860 U.S. Federal Census Manuscript Schedules," *Historical Methods: A Journal of Quantitative and Interdisciplinary History* 29, no. 4 (October 1996): 141–156, ISSN: 0161-5440, 1940-1906, doi:10.1080/01615440.1996.10112735; Jason Long and Joseph Ferrie, "Intergenerational Occupational Mobility in Great Britain and the United States Since 1850," Publisher: American Economic Association, *The American Economic Review* 103, no. 4 (2013): 1109–1137, ISSN: 00028282; Ran Abramitzky et al., "Automated Linking of Historical Data," *Journal of Economic Literature* 59, no. 3 (September 2021): 865–918, ISSN: 0022-0515, doi:10.1257/jel.20201599

7. Ran Abramitzky, Leah Platt Boustan, and Katherine Eriksson, "Europe's Tired, Poor, Huddled Masses: Self-Selection and Economic Outcomes in the Age of Mass Migration," *American Economic Review* 102, no. 5 (August 2012): 1832–1856, ISSN: 0002-8282, doi:10.1257/aer.102.5.1832; Ran Abramitzky, Leah Platt Boustan, and Katherine Eriksson, "A Nation of Immigrants: Assimilation and Economic Outcomes in the Age of Mass Migration," *Journal of Political Economy* 122, no. 3 (June 2014): 467–506, ISSN: 0022-3808, 1537-534X, doi:10.1086/675805; Ran Abramitzky, Leah Boustan, and Katherine Eriksson, "To the New World and Back Again: Return Migrants in the Age of Mass Migration," *ILR Review* 72, no. 2 (March 2019): 300–322, ISSN: 0019-7939, 2162-271X, doi:10.1177/0019793917726981

and economic change.Allen, *The British industrial revolution in global perspective*; Allen, "The Industrial Revolution in Miniature" Many of these changes were shaped by new technologies, which revolutionised the way goods were produced, traded, and consumed.[8] For many workers, especially those in labour-intensive textile work or artisanal occupations, this meant the rapid erosion of job security and a fundamental redefinition of their economic roles.[9] While traditional economic theory—particularly the Schumpeterian concept of creative destruction—frames such disruption as an inevitable and even beneficial outcome of capitalist innovation, little is known about the impact of such technological upheaval on individual workers' outcomes over the life cycle.

The impact of technological change on work, particularly for mid- and low-skill workers, remains a critical area of scholarly interest.[10] As high-income countries experience both globalisation-driven unemployment and the rapid growth of artificial intelligence, understanding historical precedent for significant occupational upheaval is especially important. By analysing the impact of technological change on workers throughout the life course, researchers can better understand how individuals respond to job loss.

This dissertation seeks to disentangle the individual experiences of economic change on workers from broader discourse on economic progress. I find that individual occupational mobility declines as workers get older, even when controlling for initial class. Further, my results show mobility is regional—certain areas are much less likely to upgrade than others. Finally, I find that workers employed in industries hit by an adverse trade shock continue bearing scars of the shock nearly twenty years after recovery, showing that short-term displacement can affect long-term employment outcomes. By tracing individual occupational trajectories over time, this research sheds light on the factors that shaped economic mobility in England and Wales from 1861 to 1911.

8. Maxine Berg, *The machinery question and the making of political economy, 1815-1848* (Cambridge, UK: Cambridge University Press, 1980), ISBN: 978-0-521-28759-3 978-0-521-22782-7; Berg, *The Age of Manufactures, 1700-1820*

9. Humphries and Schneider, "Spinning the industrial revolution"

10. Autor, Dorn, and Hanson, "The China Syndrome"; Daron Acemoglu and Pascual Restrepo, *Robots and Jobs: Evidence from US Labor Markets*, technical report w23285 (Cambridge, MA: National Bureau of Economic Research, March 2017), w23285, doi:10.3386/w23285; Rui Costa, Swati Dhingra, and Stephen Machin, *Trade and Worker Deskilling*, technical report (IZA - Institute of Labor Economics, 2019)

1.2 Research Questions

Understanding the determinants of occupational mobility is a massive endeavour, even with detailed census data. I therefore set forth two primary research questions to analyse how workers adjusted to large structural and trade-induced shocks in England and Wales, 1861–1911:

1. **How did age and regional variation affect occupational mobility in England and Wales, 1861-1911?** Using linked census data across 20-year samples, I examine general determinants of unconditional and conditional occupational mobility, with a focus on age and regional effects. This analysis situates historical evidence within contemporary debates on life-cycle adjustment and the geography of opportunity.

2. **How did the Cotton Famine (1861-65) affect long-run worker mobility?** I then examine a specific episode of trade-induced occupational decline, the Cotton Famine (1861–65). By isolating the long-run consequences of this localised shock, I analyse how displaced workers adjusted over subsequent decades. This approach parallels modern analyses of globalisation-driven trade shocks, but extends the horizon to capture life-cycle consequences of short-term industry-specific collapse.

Whereas the first question interrogates how individual and regional characteristics shape general labour market outcomes, the latter focuses on the specific individual effects of a localised, industry-specific trade shock. Together, these questions provide two important insights into worker welfare during periods of rapid economic change. This study contributes to a greater body of literature on labour market adjustment, bridging historical evidence with contemporary scholarship on labour reallocation, scarring, and the heterogeneous impacts of trade.

1.3 Roadmap

This dissertation is structured as follows:

- In Section 2, I perform a review of the literature on labour market displacement, occupational upgrading, and the cotton famine, providing a historical and economic framework of the mechanisms driving economic change in 19th-century England and Wales.

- In Section 3, I discuss the data and census linking methodology underpinning my analysis. First, I describe the processes used to link records across census samples. I then describe the data I use for occupational status indices and trade.
- In Section 4, I discuss my results on the general determinants of occupational mobility across three cohorts in England and Wales, 1861-1911. This section begins with a definition of my empirical model, followed by a presentation of my results for both unconditional and conditional mobility.
- In Section 5, I present results for worker mobility after the Cotton Famine. Mirroring the structure of the preceding section, I first outline the empirical strategy, then present results for both all workers and those employed in cotton before the shock.
- In Section 6, I provide a discussion of the results from Sections 4 and 5. I situate these results within the broader literature, evaluate possible mechanisms underlying the results, and discuss possibilities for future research.[11]
- Section 7 concludes by reflecting on the relationship between historical occupational mobility and contemporary debates on labour market adjustment.

Throughout this dissertation, I use historical data to address questions that remain central to labour economics and policy today: when the economy changes, who adapts, and who gets left behind? By leveraging data on millions of adult English and Welsh men linked across 20-year periods, the sections that follow provide insights on the life-cycle consequences of economic change to address these questions over half a century of industrial upheaval.

2 Literature

How do workers respond to rapid economic change, and how do these periods of change affect long-run labour market outcomes? These questions lie at the heart of much of modern labour economics and historical analyses of labour market change. A large body of work has emerged to address how displacement, trade shocks, and economic change impact workers in the medium and long run. In a historical context, such analyses bridge the gap of hundreds of

11. I provide robustness checks of my findings in Appendix A.

years of rapid economic change, from the Industrial Revolution to the technological revolution today. This literature review puts into conversation ongoing economic and historical debates over labour market adjustment to change. I divide this section into three subsections: the economics of displacement, the economic history of occupational mobility, and the broader historical debate over one specific trade shock, the Lancashire Cotton Famine.

2.1 Trade, Innovation, and Labour Market Displacement

While technological progress is widely regarded as the primary engine of long-run economic growth, it has historically provoked considerable anxiety regarding its disruptive effects on labour markets. Perhaps no historical phenomenon has inspired as much displacement anxiety as the Industrial Revolution.[12] The advent of new technologies and labour organisations between 1790 and 1830 led to the emergence of a new "working class," bound not by craft or ideology, but involvement in new industry.[13] By 1860, the British economy had transitioned to an industrialised state with a robust working class. Social movements quickly sprang up throughout England to resist new technologies, including through tactics such as property destruction and arson.[14] Yet, by the end of the nineteenth century, concerns over the machines had appeared to subside. Mokyr notes that such fears were still present among "the masses," but overall employment indicators suggested machines increased opportunity rather than replacing vast swaths of the workforce.[15] Berg, similarly, notes that political debates around technological unemployment in the 1830s had largely subsided only a decade later.[16]

Today, fears over automation and globalisation have once again provoked significant de-

12. Joel Mokyr, Chris Vickers, and Nicolas L. Ziebarth, "The History of Technological Anxiety and the Future of Economic Growth: Is This Time Different?," Publisher: American Economic Association, *The Journal of Economic Perspectives* 29, no. 3 (2015): p. 33, ISSN: 08953309; Jeffrey Williamson, "Earnings Inequality in Nineteenth-Century Britain," Publisher: Cambridge University Press, *The Journal of Economic History* 40, no. 3 (1980): pp. 466-67, ISSN: 00220507, 14716372; Jason Long, "The surprising social mobility of Victorian Britain," Publisher: Oxford University Press, *European Review of Economic History* 17, no. 1 (2013): p. 17, ISSN: 13614916, 14740044

13. E. P. Thompson, *The making of the English working class*, 1. Vintage ed, Vintage books History 322 (New York: Vintage Books, 1966), pp. 190-91, 194, ISBN: 978-0-394-70322-0

14. Joel Mokyr, "Technological Inertia in Economic History," Publisher: Cambridge University Press, *The Journal of Economic History* 52, no. 2 (1992): p. 330, ISSN: 00220507, 14716372

15. Mokyr, Vickers, and Ziebarth, "The History of Technological Anxiety and the Future of Economic Growth: Is This Time Different?," pp. 36-37

16. Berg, *The machinery question and the making of political economy*, pp. 129-31, 226-27

bate over what happens to workers after substantial macroeconomic change.[17] Autor, Dorn, and Hanson (henceforth, "ADH"), in their famous "China Shock" paper, find that increased exposure to Chinese manufactures reduced regional manufacturing employment.[18] ADH find that the China Shock has repercussions on household outcomes, including male mortality rates, single motherhood, and marriage rates.[19] Jobs that once offered economic security and middle-class lifestyles are disappearing, leaving affected workers with limited economic opportunity.

A longstanding view, dating back to Ricardo and Mill, treats labour displacement as a largely temporary phenomenon. Ricardo doubted the persistence of technology-driven unemployment, arguing such a problem would only occur in the long run when resources were too scarce to facilitate resource reallocation.[20] Modern empirical work largely supports Ricardo's view in aggregate: while some jobs are destroyed, innovation increases overall labour demand in new sectors.[21] In a frictionless labour market, unemployment is short-lived, as individuals rapidly respond to shifting wage incentives. At the same time, new technologies increase labour demand in two ways: through a productivity effect, which increases labour demand in tasks not replaced by the technology, and a reinstatement effect, which generates new jobs in tasks that emerge from or support the innovation.[22] From this perspective, displacement is primarily a short-run issue, as labour markets adjust to shocks over the long run.

At the same time, displaced workers face significant frictions: job search costs, retraining requirements, geographic barriers, and the social and psychological toll of occupational identity loss. A large body of work has emerged over the past three decades examining the

17. Daron Acemoglu et al., "Import Competition and the Great US Employment Sag of the 2000s," *Journal of Labor Economics* 34, no. S1 (January 2016): pp. S141-S198, ISSN: 0734-306X, 1537-5307, doi:10.1086/682384, ibid., pp. 24-28, George J. Borjas and Richard B. Freeman, "From Immigrants to Robots: The Changing Locus of Substitutes for Workers," Publisher: Russell Sage Foundation, *RSF: The Russell Sage Foundation Journal of the Social Sciences* 5, no. 5 (2019): pp. 17-18, ISSN: 23778253, 23778261, doi:10.7758/rsf.2019.5.5.02

18. Autor, Dorn, and Hanson, "The China Syndrome," p. 2139

19. Autor, Dorn, and Hanson, "When Work Disappears: Manufacturing Decline and the Falling Marriage-Market Value of Young Men," pp. 161–178

20. Berg, *The machinery question and the making of political economy*, p. 67

21. Mokyr, Vickers, and Ziebarth, "The History of Technological Anxiety and the Future of Economic Growth: Is This Time Different?," p. 36

22. Daron Acemoglu and Pascual Restrepo, "Automation and New Tasks," Publisher: American Economic Association, *The Journal of Economic Perspectives* 33, no. 2 (2019): pp. 4, 10-11, ISSN: 08953309, 19447965

worker-level effects of displacement. In their seminal paper on earnings losses after job separation, Jacobson et al. find that workers experienced depressed earnings even six years after separation, a phenomenon often referred to as "scarring.".[23] This "scarring effect" has been noted by other scholars to contribute to long-term earnings penalties.[24] ADH find that the results of their famous "China Shock" study persist through 2019, almost ten years after the initial shock reached its highest intensity.[25] In fact, recent research has found these scarring effects persist to the second generation, affecting children's schooling and earning outcomes.[26]

However, few studies examine individual worker outcomes beyond ten years post-displacement. The closest is von Wachter et al. , who use a panel of a random sample of the U.S. labour market in the late twentieth century to track earnings penalties up to two decades after job loss.[27] This dissertation builds on their approach by examining three cohorts with up to twenty-year post-shock outcomes, allowing a more comprehensive view of long-term adjustment. The limited horizon of most modern administrative data—such as that used by Jacobson et al. and Lachowska et al.—may constrain our understanding of the life-cycle effects of labour market reallocation or displacement. This raises a central unresolved question in both labour economics and economic history: do short- and medium-run scarring effects fade over the life cycle, or do they remain as long-term economic disadvantage? This dissertation attempts to answer this question by leveraging linked microdata to understand better how labour markets adjust to vast economic upheaval.

As automation and globalisation present new challenges for modern labour markets, it

23. Jacobson, LaLonde, and Sullivan, "Earnings Losses of Displaced Workers," p. 697

24. Marta Lachowska, Alexandre Mas, and Stephen A. Woodbury, "Sources of Displaced Workers' Long-Term Earnings Losses," *American Economic Review* 110, no. 10 (October 2020): p. 3231, ISSN: 0002-8282, doi:10.1257/aer.20180652, Autor, Dorn, and Hanson, "On the Persistence of the China Shock," p. 384, Martin Halla, Julia Schmieder, and Andrea Weber, "Job Displacement, Family Dynamics, and Spousal Labor Supply," Publisher: American Economic Association, *American Economic Journal: Applied Economics* 12, no. 4 (2020): p. 256, ISSN: 19457782, 19457790, David Autor et al., *Places versus People: The Ins and Outs of Labor Market Adjustment to Globalization*, technical report w33424 (Cambridge, MA: National Bureau of Economic Research, January 2025), p. 35, doi:10.3386/w33424

25. Autor, Dorn, and Hanson, "On the Persistence of the China Shock," p. 384

26. Philip Oreopoulos, Marianne Page, and Ann Huff Stevens, "The Intergenerational Effects of Worker Displacement," *Journal of Labor Economics* 26, no. 3 (July 2008): pp. 455, 468, ISSN: 0734-306X, 1537-5307, doi:10.1086/588493, Ann Huff Stevens and Jessamyn Schaller, *Short-run Effects of Parental Job Loss on Children's Academic Achievement*, technical report w15480 (Cambridge, MA: National Bureau of Economic Research, November 2009), p. 19, doi:10.3386/w15480

27. Till von Wachter, Jae Song, and Joyce Manchester, "Trends in Employment and Earnings of Allowed and Rejected Applicants to the Social Security Disability Insurance Program," *American Economic Review* 101, no. 7 (December 2011): pp. 3-7, ISSN: 0002-8282, doi:10.1257/aer.101.7.3308

is essential to understand the individual determinants of mobility, especially after an adverse shock. This dissertation contributes to the literature in two ways: first, I examine cohort-level occupational mobility from 1861 to 1911, estimating how age and regional factors affect mobility. Second, I exploit a plausibly exogenous, localised shock—the Lancashire Cotton Famine—to contribute to a growing body of literature on long-term worker outcomes following trade-driven displacement. These models contribute to our understanding of the individual-level, long-run responses to global economic change.

2.2 Occupational Mobility in Britain

While economic change often displaces workers into lower-skill jobs, it can also create new opportunities for advancement. Whether shocks lead to long-run occupational downgrading or upgrading depends on both the creation of higher-quality jobs and workers' ability to access them. A parallel literature on intergenerational mobility—measuring how occupational status is transmitted from fathers to sons and daughters—provides a powerful lens for understanding how opportunities are distributed across generations.[28]

Britain has long been viewed as having a rigid class system, especially in comparison to the United States.[29] Long and Ferrie find this to be the case until the mid-twentieth century—using linked microdata from 1851/51 through 1880/81, they find the US had both higher individual and intergenerational mobility than Britain in the nineteenth century.[30] Likewise, Clark and Cummins extend this analysis, finding incredibly high intergenerational correlations in wealth from 1858 through 2012, in an analysis of individuals with rare surnames.[31] These studies suggest a deep-rootedness in status in Britain during the period of observation.

Yet, this trend is not constrained to Britain. While Long and Ferrie find relatively high

28. For father–daughter mobility, *see* Claudia Olivetti and M. Daniele Paserman, "In the Name of the Son (and the Daughter): Intergenerational Mobility in the United States, 1850 –1940," *American Economic Review* 105, no. 8 (August 2015): pp. 2695–2724, ISSN: 0002-8282, doi:10.1257/aer.20130821.

29. Long and Ferrie, "Intergenerational Occupational Mobility in Great Britain and the United States Since 1850," p. 1111, ibid., pp. 2-3

30. ibid., pp. 1113, 1123; Long, "The surprising social mobility of Victorian Britain," pp. 16-17

31. Gregory Clark and Neil Cummins, "Intergenerational Wealth Mobility in England, 1858-2012: Surnames and Social Mobility," *The Economic Journal* 125, no. 582 (February 2015): p. 78, ISSN: 00130133, doi:10.1111/ecoj.12165

levels of mobility in the US, Abramitzky, Boustan, and Erickson (henceforth, "ABE") find immigrants to the United States experienced less occupational advancement than originally thought, a trend that persisted into the second generation.[32] Song et al., similarly, find mobility had an S-shaped path from the mid-nineteenth to late-twentieth centuries, a result echoed in Chetty et al.'s analysis of child-parent income ranks from 1996–2012.[33] These results cast doubt on the traditional narrative that Britain was far less mobile than other countries.

However, the extent to which mobility in Britain is constrained is still highly debated. For example, Clark and Cummins find very high levels of correlation between sons and fathers—between 0.68 and 0.83 in wealthier groups across generations.[34] Solon challenges Clark and Cummins' findings on intergenerational mobility, arguing this sample of rare surnames may be biased, leading to much higher coefficients (i.e. lower mobility) than other samples.[35] Other recent work incorporates grandparents' outcomes as an important indicator, suggesting status transmission beyond father-child pairings.[36] These findings, through making clear parents' outcomes matter for occupational mobility, do not necessarily examine other barriers to mobility.

This dissertation extends this literature on the barriers to mobility, asking how age, geography, and changes in local industry shaped individual labour market outcomes. While previous literature has examined status persistence across generations, this dissertation examines what factors drove occupational upgrading during the lifecycle. As such, this research situates itself within a growing body of work that seeks to disentangle the drivers of mobility,

32. Abramitzky, L. P. Boustan, and Eriksson, "Europe's Tired, Poor, Huddled Masses," o. 1842-43 Abramitzky, L. P. Boustan, and Eriksson, "A Nation of Immigrants," pp. 485, 499-500; Zachary Ward, "The Not-So-Hot Melting Pot," Publisher: American Economic Association, *American Economic Journal: Applied Economics* 12, no. 4 (2020): 73–102, ISSN: 19457782, 19457790

33. Xi Song et al., "Long-term decline in intergenerational mobility in the United States since the 1850s," Publisher: National Academy of Sciences, *Proceedings of the National Academy of Sciences of the United States of America* 117, no. 1 (2020): pp. 252-53, ISSN: 00278424, 10916490. Raj Chetty et al., "Is the United States Still a Land of Opportunity? Recent Trends in Intergenerational Mobility," *American Economic Review* 104, no. 5 (May 2014): pp. 142-44, ISSN: 0002-8282, doi:10.1257/aer.104.5.141

34. Clark and Cummins, "Intergenerational Wealth Mobility in England, 1858-2012," p. 74

35. Gary Solon, "What Do We Know So Far about Multigenerational Mobility?," Publisher: [Oxford University Press, Royal Economic Society, Wiley], *The Economic Journal* 128, no. 612 (2018): p. F348, ISSN: 00130133, 14680297

36. Zachary Ward, Kasey Buckles, and Joseph Price, *Like Great-Grandparent, Like Great-Grandchild? Multigenerational Mobility in American History*, technical report w33923 (Cambridge, MA: National Bureau of Economic Research, June 2025), pp. 25-26, doi:10.3386/w33923

using newly available linked historical data to track how individual adult male workers in England and Wales navigated structural economic change.

2.3 Industrialisation and the Lancashire Cotton Famine (1861-65)

Perhaps no industry is as often linked with labour market upheaval during British industrialisation as textile manufacturing.[37] Cotton textile manufacturing, which included the processes of spinning, weaving, bleaching, dyeing, and calico-printing, was one of the leading industries in nineteenth-century Britain, employing approximately 200,000 English factory operatives in 1834.[38] Cotton textiles remained one of the largest industries in Britain, contributing nearly one-quarter of exports on the eve of World War I.[39]

Between 1770 and 1815, different technologies, such as jennies, water frames, and mules, created demand for various skills, buildings, and raw materials.[40] By 1850, as Broadberry and Marrison argue, power looms and other technologies had already been integrated into cotton factories.[41] By mid-century, production shifted away from technological uptake toward advancements in industrial organisation. Chief among these was the shift to vertical specialisation, in which firms specialised in only one stage of the production process. Industry clustered around cotton towns in the early nineteenth century—financial services appeared to finance new business ventures, while advancements in transportation provided cheaper access to nearby coal mines and low-cost labour.[42] Broadberry and Marrison argue this "external economy of scale" allowed firms to benefit from high degrees of urban agglomeration near cotton towns in Lancashire.[43] At the same time, vertical specialisation stalled technological

37. Humphries and Schneider, "Spinning the industrial revolution"; Allen, *The British industrial revolution in global perspective*, pp. 182-216

38. M. Blaug, "The Productivity of Capital in the Lancashire Cotton Industry during the Nineteenth Century," *The Economic History Review* 13, no. 3 (1961): pp. 358-59, ISSN: 00130117, 14680289

39. Lazonick, "Industrial Organization and Technological Change: The Decline of the British Cotton Industry," p. 196

40. Stanley Chapman, "Fixed Capital Formation in the British Cotton Industry, 1770-1815," *The Economic History Review* 23, no. 2 (1970): p. 236, ISSN: 00130117, 14680289

41. Stephen Broadberry and Andrew Marrison, "External Economies of Scale in the Lancashire Cotton Industry, 1900-1950," Publisher: [Economic History Society, Wiley], *The Economic History Review* 55, no. 1 (2002): p. 55, ISSN: 0013-0117

42. Lazonick, "Industrial Organization and Technological Change: The Decline of the British Cotton Industry," p. 199

43. Broadberry and Marrison, "External Economies of Scale in the Lancashire Cotton Industry, 1900-1950," pp. 73-74

adoption, keeping cotton labour-intensive until the 1920s.[44]

While Brady argues the overproduction before the famine contributed more to the famine itself than the Civil War, there is no denying that the structure of the cotton supply chain was vulnerable to trade shocks.[45] First, Britain was entirely reliant on imports from across the world, primarily America and also Brazil, the Caribbean, and India.[46] Second, most cotton producers did keep reserves, but instead purchased supplies as needed. Mill owners closely followed cotton prices, changing production schedules according to affordability.[47] As Lazonick notes, "[The] spinning manager would buy his cotton [...] essentially from week to week, never knowing exactly what quality cotton would be available and always keeping his eyes open for feasible mixes of cotton that would enable him to cut costs."[48] As a result, Britain's cotton textile manufacturing industry was not resilient to any breakdown in the supply chain.

This weakness proved deleterious in 1861. At the onset of the US Civil War, US President Abraham Lincoln announced a blockade of Southern ports, specifically aimed at halting cotton exports. The blockade caused immediate concern in Britain, where one Member of Parliament noted that:

> [T]he distress which had been occasioned in Lancashire and in Lyons by the dearth of cotton seemed to show that the blockade was most effective. If it were true that the planters, acting under the orders of Mr. Jefferson Davis's Government, refused to sell, in the expectation of forcing the Government of this country to come to the assistance of the Southern States, he hoped no person in that House would consider himself bound to support a policy so selfish, and that the wrath of distressed operatives and sympathizing protectionists would not be directed against the Federal Government.[49]

44. Broadberry and Marrison, "External Economies of Scale in the Lancashire Cotton Industry, 1900-1950," p. 55, J.S. Toms, "The Finance and Growth of the Lancashire Cotton Textile Industry, 1870-1914," Publisher: Cambridge University Press, *Business and Economic History* 26, no. 2 (1997): p. 327, ISSN: 08946825

45. Eugene Brady, "A Reconsideration of the Lancashire "Cotton Famine"," Publisher: Agricultural History Society, *Agricultural History* 37, no. 3 (1963): p. 75, ISSN: 0002-1482

46. Raw cotton is not a commodity, but varies significantly by source countries. Quality differences are reflected in the prices of cotton. W. Walker Hanlon, "Necessity Is the Mother of Invention: Input Supplies and Directed Technical Change: Directed Technical Change," *Econometrica* 83, no. 1 (January 2015): p.75-76, ISSN: 00129682, doi:10.3982/ECTA10811

47. Theo Balderston, "The economics of abundance: coal and cotton in Lancashire and the world," Publisher: [Economic History Society, Wiley], *The Economic History Review* 63, no. 3 (2010): p. 571, ISSN: 00130117, 14680289

48. Lazonick, "Industrial Organization and Technological Change: The Decline of the British Cotton Industry," p. 208

49. House of Commons, *United States—Blockade of the Southern Ports*, Hansard, vol. 165, cc. 526–532, UK Parliament, February 1862

Between 1861 and 1865, American cotton exports to Britain fell by 87.8 per cent, resulting in the so-called "Lancashire Cotton Famine."[50] This particular trade shock came only one year after the signing of the Cobden-Chevalier Treaty. As such, the expected gains from liberalisation were immediately stifled by the shock to Britain's access to raw cotton.[51]

The economic devastation of the Cotton Famine has been well documented in both primary and secondary literature.[52] Workers turned to governmental and charitable assistance, or poor relief, to offset lost wages. At the height of the Famine, nearly one in three people in some Poor Law unions were receiving assistance.[53] By 1862, Lancashire reached unprecedented levels of unemployment. In Preston in May 1862, over eight per cent of the population were paupers, and the percentage of the population receiving poor relief rose from 2.9 per cent in September 1861 to 21 per cent by November 1862.[54] In areas with mines or stone yards, workers sought employment in this physically demanding work, often with little to no training.[55] Workers faced famine, prompting riots from Stalybridge, Prescott, Manchester, and Wigan in 1862.[56]

The plausibly exogenous nature of the cotton shock, combined with its highly localised impact, allows researchers to isolate the effects of this severe supply-side trade shock on employment. However, no study has yet examined the long-term labour market consequences of the Cotton Famine. Arthi et al. isolate cohorts living in exposed regions in 1861 and link these records with death records, finding the Cotton Famine had adverse effects on affected individuals' long-term health outcomes, as well as mortality outcomes in their households.[57]

50. Brian Mitchell and Phyllis Deane, *Abstract of British Historical Statistics*, Reprinted, Monographs / University of Cambridge, Department of Applied Economics 17 (Cambridge: Cambridge University Press, 1962), pp. 180-81, ISBN: 978-0-521-05738-7. $\frac{1{,}116-136}{1{,}116} = 0.8781362$.

51. D. J. Oddy, "Urban Famine in Nineteenth-Century Britain: The Effect of the Landcashire Cotton Famine on Working-Class Diet and Health," Publisher: [Economic History Society, Wiley], *The Economic History Review* 36, no. 1 (1983): p. 72, ISSN: 0013-0117, doi:10.2307/2598898, Brady, "A Reconsideration of the Lancashire "Cotton Famine"," p. 156

52. Rosalind Hall, "A Poor Cotton Weyver: Poverty and the Cotton Famine in Clitheroe," Publisher: Taylor & Francis, Ltd. *Social History* 28, no. 2 (2003): p. 288, ISSN: 03071022

53. George Boyer, "The Evolution of Unemployment Relief in Great Britain," *The Journal of Interdisciplinary History* 34, no. 3 (2004): p. 404, ISSN: 00221953, 15309169

54. Oddy, "Urban Famine in Nineteenth-Century Britain," p. 72, Boyer, "The Evolution of Unemployment Relief in Great Britain," p. 405

55. Oddy, "Urban Famine in Nineteenth-Century Britain," p. 75

56. ibid.

57. Vellore Arthi, Brian Beach, and W. Walker Hanlon, "Recessions, Mortality, and Migration Bias: Evidence from the Lancashire Cotton Famine," *American Economic Journal: Applied Economics* 14, no. 2 (April 2022): pp. 230-31, ISSN: 1945-7782, 1945-7790, doi:10.1257/app.20190131

Hanlon uses the Cotton Famine trade shock to explore the impact of changes in input prices on technological progress.[58] Oddy explores the impact of the Cotton Famine on workers' diets, finding that nutritional deficiencies due to reduced wages led to famine across affected towns.[59]

This dissertation builds on these projects by using the cotton trade shock to investigate long-term worker mobility. The Lancashire Cotton Famine provides a compelling natural experiment: decreased American cotton production during the Civil War caused a large, highly localised, and plausibly exogenous shock to British textile production. By examining worker outcomes before the shock (1861) and after the shock (1881), this study tracks the long-term effects on worker displacement and mobility.

3 Data and Research Methodology

This section describes the data cleaning used to produce linked census records of adult males from 1861 to 1911. My analysis relies on the Integrated Census Microdata (I-CeM), which provide tens of millions of standardised records, including demographics, occupation, and location.[60] To supplement these data, I also describe the use of occupational status indices, namely HISCAM-GB, which allow me to measure changing occupational status over time.[61] Finally, I incorporate trade data on raw cotton imports from the US and elsewhere to analyse the Cotton Famine. Combining these sources produces precise, linked longitudinal samples for studying individual labour market adjustments across England and Wales.

58. Hanlon, "Necessity Is the Mother of Invention," pp. 67-68

59. Oddy, "Urban Famine in Nineteenth-Century Britain," pp. 83-84

60. K. Schurer et al., *Integrated Census Microdata (I-CeM), 1851-1911*, 2025, doi:10.5255/UKDA-SN-7481-3; K. Schurer and E. Higgs, *Integrated Census Microdata (I-CeM) Names and Addresses, 1851-1911: Special Licence Access*, 2025, doi:10.5255/UKDA-SN-7856-2

61. Paul S. Lambert et al., "The Construction of HISCAM: A Stratification Scale Based on Social Interactions for Historical Comparative Research," *Historical Methods: A Journal of Quantitative and Interdisciplinary History* 46, no. 2 (April 2013): pp. 77-89, ISSN: 0161-5440, 1940-1906, doi:10.1080/01615440.2012.715569

3.1 Integrated Census Microdata (I-CeM)

When Britain began taking decennial censuses beginning in 1801, participants were asked whether they worked in agriculture, a "trade, manufacture or handicraft," or "other."[62] Early censuses were hindered by poor record-keeping and oversight.[63] These limitations were reduced in the 1830s, spurred by Benthamite reforms, which paved the way for more systematic and accurate census administration.[64]

While Higgs argues that errors in census records after 1841 were relatively small, issues could still arise from human measurement error, recording inconsistencies, or lost records.[65] Further, the census was not originally intended to record occupational information. Instead, the Victorian General Register Office (GRO) sought to collect health records, including total deaths.[66] Therefore, census records were not recorded to capture detailed economic details, and lack information on educational status, literacy, industry, and other factors that are standard in modern labour economics.

Despite these limitations, British census records provide highly detailed accounts of individual lives across England and Wales from 1851-1911. They include rich demographic and occupational detail, along with highly granular birthplace information recorded down to the parish level. I exploit this granularity to link individuals across twenty-year cohorts, constructing new evidence on occupational mobility and adjustment to economic shocks.

3.1.1 Linking Methodology

Researchers across economics, public health, and political science have long used record-linking techniques to track individual outcomes over time.[67] Unlike cohort-level analyses,

62. Edward Higgs et al., *Integrated Census Microdata (I-CeM) Guide*, technical report (University of Essex, September 2013), p. 6

63. A. J. Taylor, "Taking of the Census, 1801-1951," *British Medical Journal* 1, no. 4709 (April 1951): p. 716, ISSN: 0007-1447, doi:10.1136/bmj.1.4709.715

64. ibid.

65. Higgs et al., *Integrated Census Microdata (I-CeM) Guide*, pp. 22-23

66. Edward Higgs, "Occupational Censuses and the Agricultural Workforce in Victorian England and Wales," *The Economic History Review* 48, no. 4 (November 1995): p. 702, ISSN: 00130117, doi:10.2307/2598131

67. Ferrie, "A New Sample of Males Linked from the Public Use Microdata Sample of the 1850 U.S. Federal Census of Population to the 1860 U.S. Federal Census Manuscript Schedules," pp. 141-156, Abramitzky, L. P. Boustan, and Eriksson, "Europe's Tired, Poor, Huddled Masses," 1832–56, Long and Ferrie, "Intergenerational Occupational Mobility in Great Britain and the United States Since 1850," pp. 1109-1137, William J. Collins and Ariell Zimran, "Working Their Way Up? US Immigrants' Changing Labor Market

which can obscure within-cohort differences, census-linking allows researchers to examine individual-level trends over the life cycle. To identify a sample of individuals over time, researchers often identify "features," such as name, sex, age, race, and place of birth, that are time-invariant. Census-linking is particularly valuable when studying occupational change and decline, as workers' job attachment and experience evolve as they age. By tracking the same individuals across decades, researchers can distinguish between short-term displacement and persistent scarring, and can examine how characteristics such as age and geography shape workers' mobility.

Still, census linking comes with methodological challenges, namely minimising false matches (Type 1) and missed matches (Type 2). To minimise missed matches, researchers standardise nicknames, use phonetic spellings of names, and relax restrictions on exact age matches to address age heaping.[68] Even with perfect matching procedures, some records are impossible to link due to name changes, emigration, or other factors. False positives are particularly concerning, as they can bias estimates.[69] To minimise false positives, researchers validate matches on other time-invariant characteristics, such as parents' features, or use strict match criteria, such as individuals with rare surnames.[70] Recent work has further refined these approaches using computational and machine-learning techniques to enhance linkage rates.[71]

Name standardisation is a key challenge in matching methodology. Limiting analysis to uncommon names, as in Ferrie's seminal work, can bias estimates when common names correlate with region or class.[72] Researchers can, instead, standardise names using phonetic spellings, such as Soundex or New York State Identification and Intelligence System

Assimilation in the Age of Mass Migration," *American Economic Journal: Applied Economics* 15, no. 3 (July 2023): pp. 238–69, ISSN: 1945-7782, 1945-7790, doi:10.1257/app.20210008, Martha Bailey et al., "How Well Do Automated Linking Methods Perform? Lessons from US Historical Data," *Journal of Economic Literature* 58, no. 4 (December 2020): pp. 997-1044, ISSN: 0022-0515, doi:10.1257/jel.20191526, Ward, Buckles, and Price, *Like Great-Grandparent, Like Great-Grandchild?*, pp. 1-63

68. Abramitzky et al., "Automated Linking of Historical Data," pp. 871-73

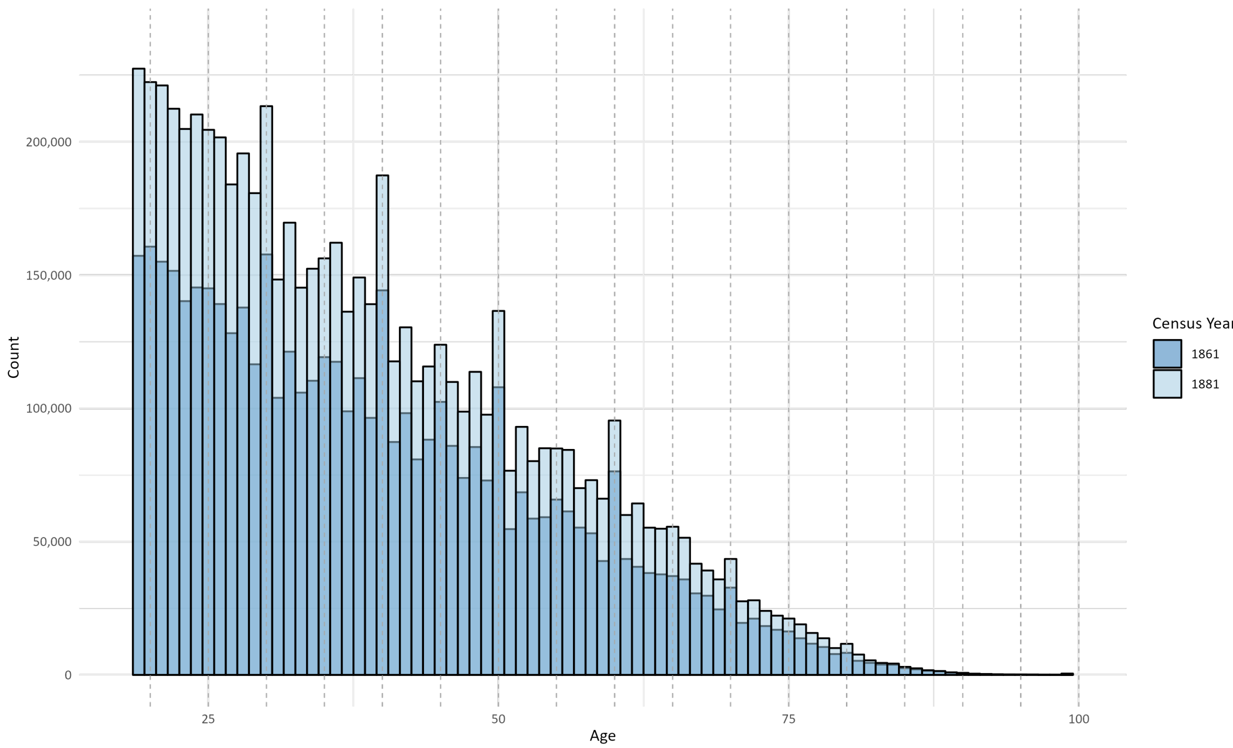
69. Cormac Ó Gráda et al., "The problem of false positives in automated census linking: Nineteenth-century New York's Irish immigrants as a case study," *Historical Methods: A Journal of Quantitative and Interdisciplinary History* 56, no. 4 (October 2023): pp. 240-259, ISSN: 0161-5440, 1940-1906, doi:10.1080/01615440.2024.2312293

70. Clark and Cummins, "Intergenerational Wealth Mobility in England, 1858-2012," p. 65

71. Abramitzky et al., "Automated Linking of Historical Data," pp. 875-78

72. Ferrie, "A New Sample of Males Linked from the Public Use Microdata Sample of the 1850 U.S. Federal Census of Population to the 1860 U.S. Federal Census Manuscript Schedules," pp. 12-15

Figure 1: Age Heaping in the 1861-1881 Censuses



Notes: K. Schurer et al., *Integrated Census Microdata (I-CeM), 1851-1911*, 2025, doi:10.5255/UKDA-SN-7481-3; K. Schurer and E. Higgs, *Integrated Census Microdata (I-CeM) Names and Addresses, 1851-1911: Special Licence Access*, 2025, doi:10.5255/UKDA-SN-7856-2

(NYSIIS).[73] However, phonetic standardisation can reduce matches for common names by homogenising distinct spellings (e.g., "JOHN SMYTHE" vs. "JON SMITH"). To reduce this error, I follow the ABE-JW method, which utilises Jaro-Winkler name similarity scores to measure string distance, with a cutoff of 10% dissimilarity.[74]

3.1.2 Data Construction

I use an adapted ABE-JW record-linking approach to match adult men across three census samples: 1861–1881, 1881–1901, and 1891–1911. The methodology follows a deterministic procedure outlined below:

1. I first clean the relevant annual samples, merging non-anonymised and anonymised

73. Bailey et al., "How Well Do Automated Linking Methods Perform?," p. 7

74. For example, "SMYTHE" and "SMITH" differ by two letters, and have a Jaro-Winkler score of ≈0.142. Abramitzky et al., "Automated Linking of Historical Data," p. 872

samples using unique record IDs. At this stage, I remove individuals who do not report any occupation, and limit my sample to adult males.

2. I then clean and standardise all first and last names. After removing any spaces or special characters from first and last names, I adjust all nicknames noted in the ABE methodology.[75]
3. After cleaning the data, I block the sample on birthplace, and first and last initial. This process both decreases computational load and rates of false positives, as individuals are only compared to others within their birthplace parish and with the same first and last initial.
4. I then begin to link these blocks. Within each blocked sample, I check if there are any exact matches within one year of birth.
   (a) If there are multiple exact matches on name and year of birth, I drop this individual ID from my sample.
   (a) If there is one exact match on name and year of birth, I select this match and move on to the next individual.
5. If there are no exact matches, I pull the match that has the closest year of birth and Jaro-Winkler distance.
   (a) If there is one unique match within the 10% Jaro-Winkler similarity and ± 1 year of birth, I select this one observation.
   (b) If there are multiple matches within the Jaro-Winkler and age cut-off, I move to the next record.
6. I then append these blocked samples into a fully linked sample for each set of years.

3.1.3 Summary Statistics

My final sample includes 4,441,427 linked census records of adult men in England and Wales across three samples: 1861-81, 1881-1901, and 1891-1911. I report linkage rates in Table 1. Across all three samples, I achieve a linkage rate between 17 and 26 per cent. These relatively low rates may stem from my tight selection criteria—whereas much of the literature relies on 5-year age bands, my analysis is restricted to a 2-year age band, which only corrects for

75. Abramitzky et al., "Automated Linking of Historical Data," p. 8

possible issues with birthdays and reporting. However, considering the large resulting sample size and the high rate of false positives created upon loosening criteria, my final sample is large enough to track individual outcomes.

Table 1: Match Rates by Census Period

Pair	Raw	Linked	Ratio
1861–1881	5,026,971	865,592	17.22%
1881–1901	6,954,527	1,541,872	22.17%
1891–1911	7,680,420	2,033,963	26.48%

Notes: Schurer et al., *Integrated Census Microdata (I-CeM), 1851-1911*; Schurer and Higgs, *Integrated Census Microdata (I-CeM) Names and Addresses, 1851-1911*

Predictably, match rates declined throughout the life course, as shown in Table 2

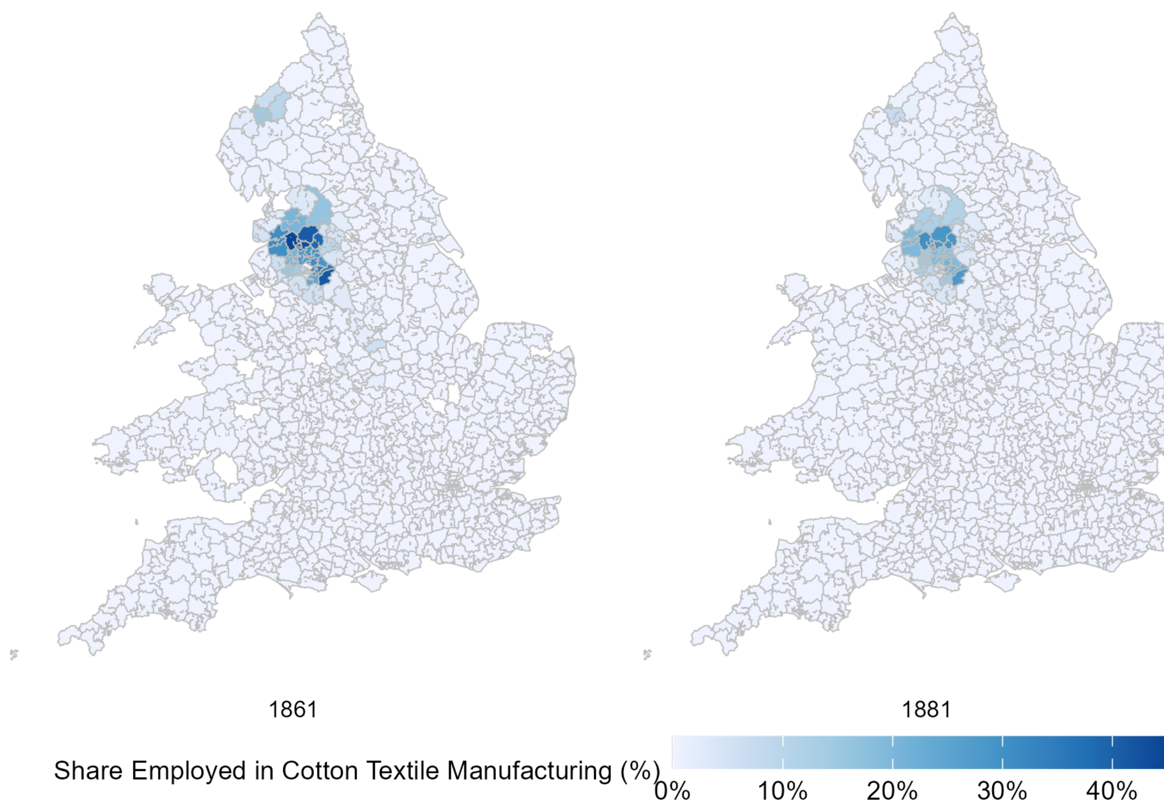
Table 2: Match Rates by Census Period and Age Group

Pair	Age Range	Raw	Linked	Ratio
1861–1881	18–25	1,216,134	266,289	21.90%
1861–1881	25–35	1,240,293	260,779	21.03%
1861–1881	35–45	1,025,613	199,542	19.46%
1861–1881	45–55	732,997	105,696	14.42%
1861–1881	55–65	485,844	30,866	6.35%
1861–1881	66+	326,090	2,420	0.74%
1881–1901	18–25	1,741,330	499,155	28.67%
1881–1901	25–35	1,747,631	478,437	27.38%
1881–1901	35–45	1,371,896	335,847	24.48%
1881–1901	45–55	976,459	172,425	17.66%
1881–1901	55–65	679,055	51,955	7.65%
1881–1901	66+	438,156	4,053	0.93%
1891–1911	18–25	1,932,909	674,791	34.91%
1891–1911	25–35	1,970,602	646,239	32.79%
1891–1911	35–45	1,518,072	424,596	27.97%
1891–1911	45–55	1,097,929	222,686	20.28%
1891–1911	55–65	698,215	59,902	8.58%
1891–1911	66+	462,693	5,749	1.24%

Notes: Schurer et al., *Integrated Census Microdata (I-CeM), 1851-1911*; Schurer and Higgs, *Integrated Census Microdata (I-CeM) Names and Addresses, 1851-1911*

Since linked data capture a subset of the full population, and are therefore potentially biased toward certain individuals, I construct inverse-propensity scores designed to reweight the linked sample. I calculate these weights by first calculating a probit model of linkage on variables used to construct linkage (binary), namely the presence of a middle name, first, middle, and last name length, name commonness, and household size.[76] All regression analyses incorporate these weights, trimmed to the 95th percentile.[77]

Figure 2: Share of Workforce Employed in Cotton Textile Manufacturing



Notes: Schurer et al., *Integrated Census Microdata (I-CeM), 1851-1911*; Schurer and Higgs, *Integrated Census Microdata (I-CeM) Names and Addresses, 1851-1911*. GIS boundary files were provided by Max Satchell, and were constructed from parish files by Joe Day for the ESRC fertility project directed by Alice Reid. A.E.M Satchell et al., *1851 England and Wales census parishes, townships and places*, 2016, doi:10.5255/UKDA-SN-852232.

3.2 Occupational Data

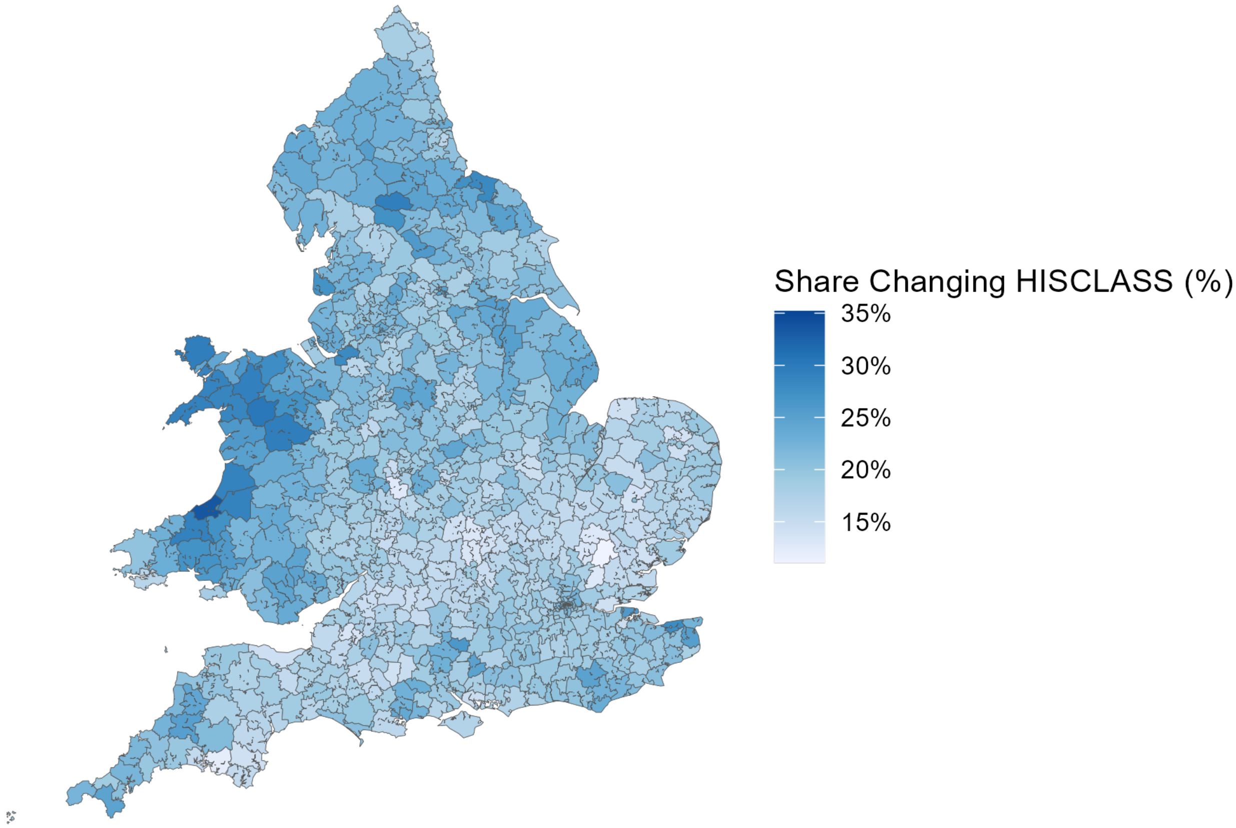
Since historical data do not provide information on wage rates or average earnings, researchers often use occupations as a proxy for economic status and well-being.[78] The I-CeM database includes two standardised variables to identify occupation: Historical International Standard

76. Bailey et al., "How Well Do Automated Linking Methods Perform?," p. 1035

77. I trim the sample using the weights to ensure names that are particularly difficult to link do not bias estimates. ibid.

78. Abramitzky et al., "Automated Linking of Historical Data," p. 895

Figure 3: Percentage of Workforce Changing HISCLASS, 1861-81



Notes: Schurer et al., *Integrated Census Microdata (I-CeM), 1851-1911*; Schurer and Higgs, *Integrated Census Microdata (I-CeM) Names and Addresses, 1851-1911*; Marco HD Van Leeuwen and Maas, *HISCLASS: A historical international social class scheme*; Lambert et al., "The Construction of HISCAM," pp. 77-89. GIS boundary files were provided by Max Satchell, and were constructed from parish files by Joe Day for the ESRC fertility project directed by Alice Reid. Satchell et al., *1851 England and Wales census parishes, townships and places*.

of Classification of Occupations (HISCO) and *OCCODE*, which was built using 1911 data based on Hollerith codes. Status indices are generally built off standardised occupational lists, such as HISCO.[79] Clark, Cummins, and Curtis, for example, use historical data on literacy, probate rates, wealth, education, and labour force participation to define occupational status indices.[80] This analysis relies upon Historical Cambridge Social Interaction and Stratification (HISCAM) scores, which were developed using data on British intergenerational mobility

79. Marco H.D. Van Leeuwen and Ineke Maas, "Historical Studies of Social Mobility and Stratification," *Annual Review of Sociology* 36, no. 1 (June 2010): 429–451, ISSN: 0360-0572, 1545-2115, doi:10.1146/annurev.soc.012809.102635

80. Gregory Clark, Neil Cummins, and Matthew Curtis, "Three new occupational status indices for England and Wales, 1800–1939," *Historical Methods: A Journal of Quantitative and Interdisciplinary History* 57, no. 1 (January 2024): p. 45, ISSN: 0161-5440, 1940-1906, doi:10.1080/01615440.2024.2368458

from 1800 to 1938.[81] To measure social class over time, I rely on a simplified version of HISCLASS, which is standard in historical analyses of British census data.[82]

Yet, this classification is limited. Since the GRO was focused on health outcomes, it instructed enumerators to note the material with which individuals worked (e.g., coal, wood, cotton, etc.), rather than the tasks they performed. Those without a specialisation in specific materials were often described as "general labourers," more of a catch-all than a classification.[83] As shown in Table 3, individuals inconsistently described their tasks. As such, occupational titles from census data do not perfectly describe the task or duration of the work itself.[84]

Table 3: Occupation Strings for Sample of HISCO 75000 ("Textile workers, specialization unknown")

HISCO	OCCODE	OCCODE Description	Occupation String
75000	565	other weaving processes (wool)	OVERLOOKER OF CLOTH MILL
75000	566	woollen cloth manufacture other processes	BLANKET RAISER
75000	572	worsted and stuff manufacture undefined	WORSTED MILL AND OVERLOOKER
75000	572	worsted and stuff manufacture undefined	BOOT KEEPER WORSTED FACTORY
75000	572	worsted and stuff manufacture undefined	OVERLOOKER COMBING MACHANIC
75000	578	silk workers - other processes	SILK PATTERN DESIGNER
75000	579	silk workers - undefined	SILK DRESSER
75000	580	flax linen and damask manufacture (various)	LINEN WEAVER
75000	580	flax linen and damask manufacture (various)	HANDLOOM LINEN WEAVER
75000	580	flax linen and damask manufacture (various)	LINEN WAREHOUSEMAN
75000	580	flax linen and damask manufacture (various)	LINEN HAND LOOM WEAVER
75000	580	flax linen and damask manufacture (various)	POWERLOOM LINEN WEAVER
75000	596	carpet rug manufacture (various)	STEAM POWER CARPET BOBBIN TURNOR
75000	608	factory hands (textile) undefined (various)	FANCY HAND LOOM WEAVER

Notes: Schurer et al., *Integrated Census Microdata (I-CeM), 1851-1911*; Schurer and Higgs, *Integrated Census Microdata (I-CeM) Names and Addresses, 1851-1911*

I preserve the *OCCODE* variable in my analysis, since it provides more granular task descriptions, as seen in Table 4. I identify cotton workers using a string match of *OCCODE*

81. All data used rely on HISCAM-GB version 1.3.1.U1. Lambert et al., "The Construction of HISCAM," pp. 77-89. Where status values are missing for certain HISCO values, I manually match the data using context from both HISCO and *OCCODE* descriptions to find the best fit. I am not able to manually identify comparable statuses for all occupations in the data, but I do get coverage for most of the occupations present in the data using this method. Appendix B for details on this process.

82. Marco HD Van Leeuwen and Ineke Maas, *HISCLASS: A historical international social class scheme* (Universitaire Pers Leuven, 2011). I sort HISCLASS into simplified groups based on Long and Ferrie, "Intergenerational Occupational Mobility in Great Britain and the United States Since 1850," p. 1113. Appendix B.

83. Higgs, "Occupational Censuses and the Agricultural Workforce in Victorian England and Wales," p. 702

84. While this analysis is limited to adult males, this problem is especially pronounced among women, who were sometimes given the same occupation as their husbands. ibid., p. 704

descriptions, searching for the words "fustian," "muslin," and, of course, "cotton." This captures all occupation codes that directly relate to cotton, where specified.

Table 4: OCCODE and Description for HISCO 75000 ("Textile workers, specialization unknown")

HISCO	OCCODE	OCCODE Description
75000	563	Flannel manufacture (various)
75000	564	Blanket manufacture (various)
75000	565	Other weaving processes (wool)
75000	566	Woollen cloth manufacture other processes
75000	567	Tartan and wincey manufacture
75000	568	Worsted and stuff manufacture other processes
75000	571	Woollen cloth manufacture undefined
75000	572	Worsted and stuff manufacture undefined
75000	575	Crepe gauze manufacture (various)
75000	578	Silk workers - other processes
75000	579	Silk workers - undefined
75000	580	Flax linen and damask manufacture (various)
75000	581	Hemp manufacture (various)
75000	582	Jute manufacture (various)
75000	583	Cocoa fibre manufacture (various)
75000	585	Mat manufacture (various)
75000	591	Thread manufacture (various)
75000	596	Carpet rug manufacture (various)
75000	607	Other workers sundry fabrics undefined
75000	608	Factory hands (textile) undefined (various)

Notes: Schurer et al., *Integrated Census Microdata (I-CeM), 1851-1911*; Schurer and Higgs, *Integrated Census Microdata (I-CeM) Names and Addresses, 1851-1911*

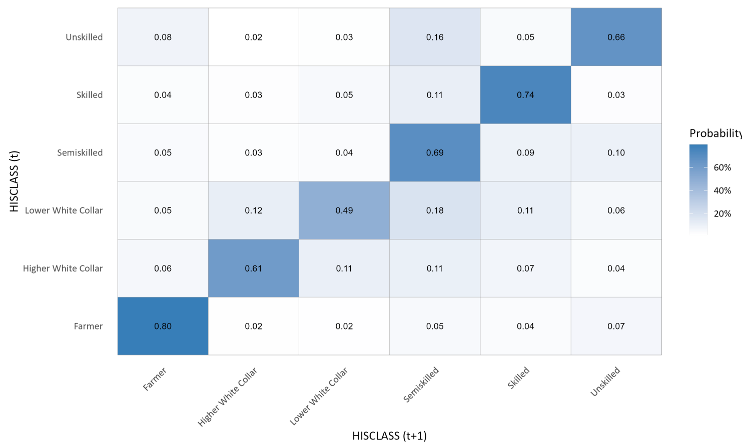
3.2.1 Measuring Occupational Change

A key question in labour economics is what drives employment reallocation. Since David Lilien's 1982 study on the issue, a vast body of literature has explored how to disentangle sectoral shifts and changes in aggregate demand.[85] This question remains central in modern

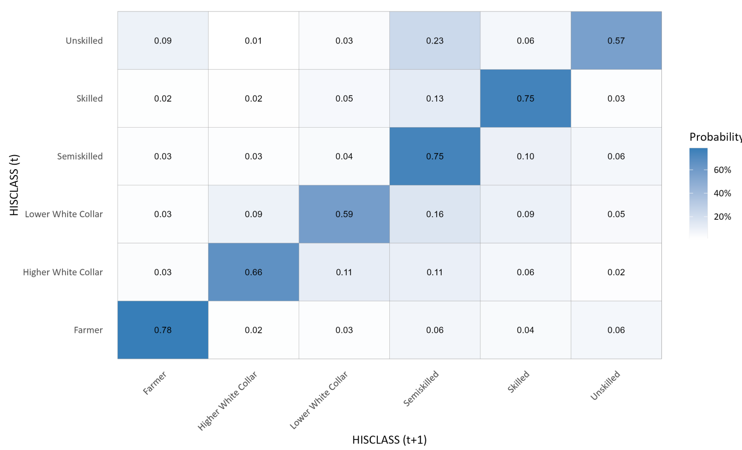
85. David M. Lilien, "Sectoral Shifts and Cyclical Unemployment," Publisher: University of Chicago Press, *Journal of Political Economy* 90, no. 4 (1982): pp. 777-793, ISSN: 00223808, 1537534X, Katharine G. Abraham and Lawrence F. Katz, "Cyclical Unemployment: Sectoral Shifts or Aggregate Disturbances?," Publisher: University of Chicago Press, *Journal of Political Economy* 94, no. 3 (1986): pp. 507-522, ISSN: 00223808, 1537534X, Gabriel Chodorow-Reich and Johannes Wieland, "Secular Labor Reallocation and Business Cycles," *Journal of Political Economy* 128, no. 6 (June 2020): pp. 2245-87, ISSN: 0022-3808, 1537-534X, doi:10.1086/705717

Figure 4: Transition Matrices by Simplified HISCLASS, 1861-1911

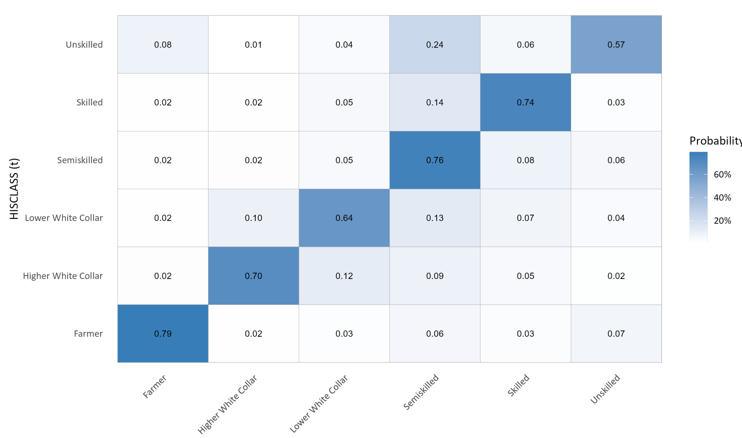
(a) 1861-81



(b) 1881-1901



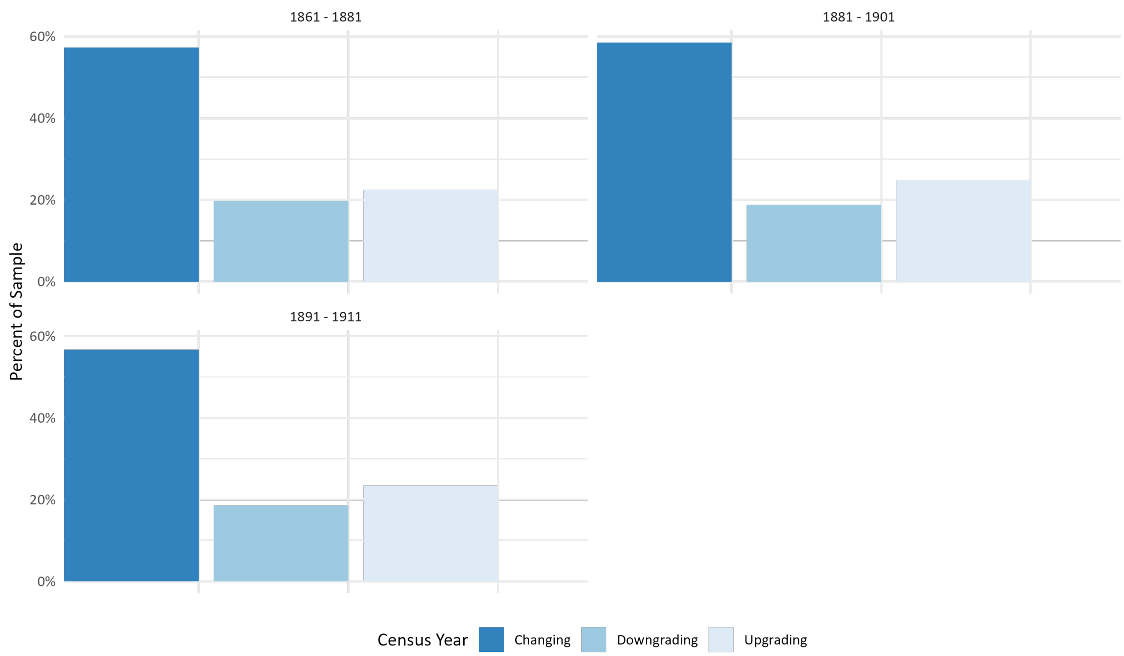
(c) 1891-1911



Notes: Schurer et al., *Integrated Census Microdata (I-CeM), 1851-1911*; Schurer and Higgs, *Integrated Census Microdata (I-CeM) Names and Addresses, 1851-1911*; Marco HD Van Leeuwen and Maas, *HISCLASS: A historical international social class scheme* labour market dynamics, as concerns over automation and globalisation raise new questions about long-term unemployment.

British census data before 1911 lack industry classifications, making traditional sectoral analyses unreliable or prone to substantial measurement error. As such, this dissertation measures labour reallocation more directly—by examining whether workers experienced occupational mobility across 20-year samples. I therefore construct a binary variable capturing whether individuals changed occupations, and whether these changes involved a transition to higher-status occupations.[86]

Figure 5: Per cent of Workers Changing, Downgrading, and Upgrading Occupations



Notes: Schurer et al., *Integrated Census Microdata (I-CeM), 1851-1911*; Schurer and Higgs, *Integrated Census Microdata (I-CeM) Names and Addresses, 1851-1911*; Lambert et al., "The Construction of HISCAM," pp. 77-89

I focus on the binary upgrading, instead of a multinomial model of upgrading and downgrading, due to the rise of industrial and clerical roles during the late nineteenth century. Workers who are already in low-status jobs are less likely to find comparable lower-skill work, whereas there is large growth in more mid- or comparable low-skill work during this period. In my empirical analysis, my primary focus is on low-skill workers, namely cotton textile work-

86. Marco HD Van Leeuwen and Maas, *HISCLASS: A historical international social class scheme*; Lambert et al., "The Construction of HISCAM," pp. 77-89

ers. These occupations are considered "low status." For example, the CCC-HISCO index, developed by Clark, Cummins, and Curtis, assigns "Bleachers, Dyers, and Textile Product Finishers" a score of 43.0, whereas HISCAM-GB assigns these a 50.95.[87] As such, I focus on upgrading rather than downgrading or occupational reallocation more generally.

3.3 Trade Data

Cotton was the heart of British industrialisation.[88] Britain's early entrance into the cotton trade through colonial networks ensured a steady supply of cotton to its mills.[89] This advantage proved significant—cotton textiles remained the largest part of world trade throughout the nineteenth century.[90]

Trade data on British imports of American raw cotton are sourced from Mitchell and Deane.[91] These data include both average prices of cotton from "Upland or Midling America," raw cotton imports from the United States and total British cotton imports. As shown in Figure 6, British imports of cotton rapidly grew throughout the nineteenth century. Until 1861, American cotton made up nearly all of British cotton imports.[92] However, American cotton imports dropped nearly to zero during the period of the US Civil War (1861-65).

While cotton was a localised trade, regional and local trade data are not reliably available across regions for the period of observation. To estimate the size of the trade shock, I measure the relative shock in American cotton imports during the affected period (1861-65) compared to cotton imports from the rest of the world (ROW) in that period. First, I calculate the pre-shock share (s) of raw cotton imports M from pre-shock year 1860 for both the US and ROW:

$$\Delta s_{1860}^{US} = \frac{M_{1860}^{US}}{M_{1860}^{ROW} + M_{1860}^{US}}; \Delta s_{1860}^{ROW} = 1 - \Delta s_{1860}^{US} \tag{1}$$

I then define the aggregate trade shock as the weighted relative log change in cotton

87. Clark, Cummins, and Curtis, "Three new occupational status indices for England and Wales, 1800–1939"; Lambert et al., "The Construction of HISCAM," pp. 77-89
88. Allen, *The British industrial revolution in global perspective*, pp. 182-216
89. Hobsbawm, *The age of revolution*, pp. 33-36
90. Berg, *The Age of Manufactures, 1700-1820*, p. 14
91. Mitchell and Deane, *Abstract of British Historical Statistics*, pp. 180-81
92. Figure 7.

imports between the US and the ROW:

$$g^{UK}_{1861-65} = s^{US}_{1860} \times log(\Delta M^{US}_{1861-65}) - s^{ROW}_{1860} \times log(\Delta M^{ROW}_{1861-65}) \tag{2}$$

Where $\Delta M^{j}_{1861-65}$ is defined as the change in imports from 1861-1865 (the period of the Cotton Famine) for the US and the rest of the world. This equation weights the size of the shock by pre-shock dependence on American cotton, and reduces the shock by substitution to non-US suppliers.

However, this only provides a shock estimate for the whole of the United Kingdom. To measure the impact of these shocks by region, I multiply the shock by the regional employment share in cotton textile manufacturing in 1861. Notably, the 1861 census was recorded before the onset of the Civil War, making these numbers exogenous to the shock.[93] For each local region r, I calculate the share of labour (θ) in registration district r employed in cotton textile manufacturing (c) as:

$$\theta^{UK}_{r,c} = \frac{L_{r,c,1861}}{\sum L_{r,1861}} \tag{3}$$

The regional impact of the trade shock is therefore calculated as:

$$\text{CottonShock}^{UK}_{r,c,1861-65} = \theta^{UK}_{r,c} \times g^{UK}_{1861-65} \tag{4}$$

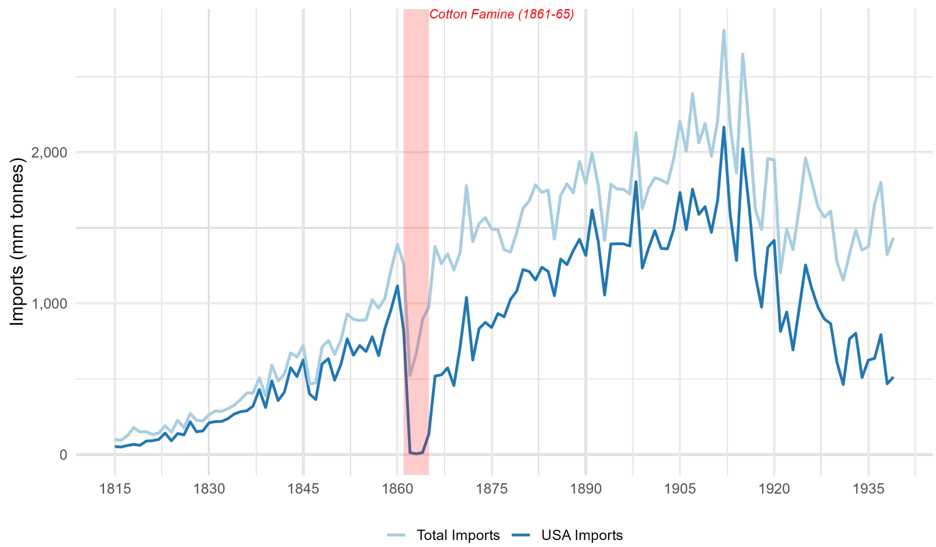
This measure captures the regional burden of the shock to UK cotton during the Cotton Famine.

4 General Determinants of Occupational Upgrading

A central question in labour economics is what enables occupational mobility. Occupational mobility, both within and across generations, has repercussions on economic development, inequality, and social cohesion. The primary research question of this paper is the determinants of occupational mobility during periods of rapid economic change. As such, I first estimate how various factors, namely age and region, impact mobility. This section examines

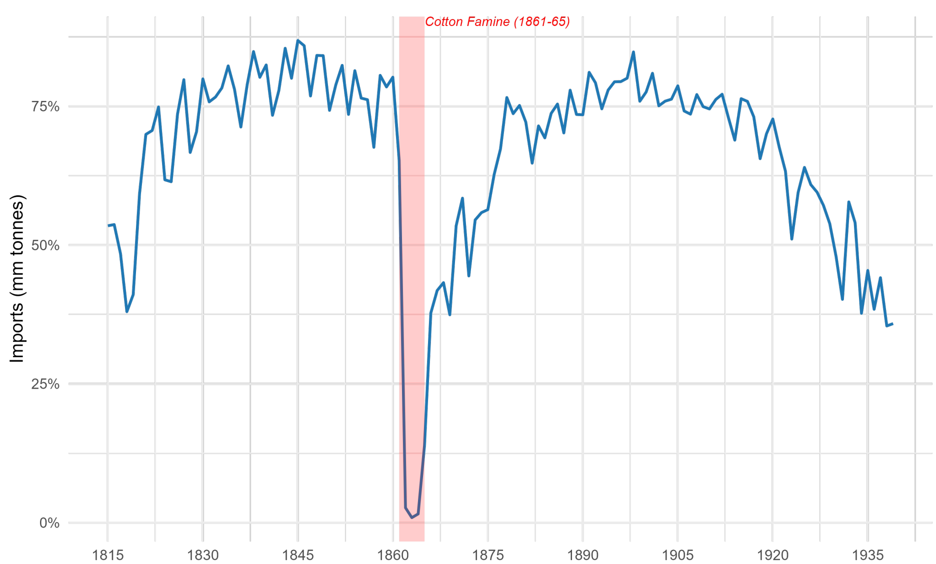
93. Higgs et al., *Integrated Census Microdata (I-CeM) Guide*, p. 40

Figure 6: Volume of Raw Cotton Imports to the United Kingdom, 1815-1938



Notes: Brian Mitchell and Phyllis Deane, *Abstract of British Historical Statistics*, Reprinted, Monographs / University of Cambridge, Department of Applied Economics 17 (Cambridge: Cambridge University Press, 1962), pp. 180-81, ISBN: 978-0-521-05738-7

Figure 7: Proportion of American to Total Raw Cotton Imports to the United Kingdom, 1815-1938



Notes: Brian Mitchell and Phyllis Deane, *Abstract of British Historical Statistics*, Reprinted, Monographs / University of Cambridge, Department of Applied Economics 17 (Cambridge: Cambridge University Press, 1962), p.180-81, ISBN: 978-0-521-05738-7

the general determinants of occupational mobility across cohorts. Using the empirical models outlined below, I highlight key patterns in both unconditional and conditional occupational upgrading.

The analysis yields several broad findings:

- **Age and Life-Cycle Effects:** The likelihood of occupational upgrading decreases over the life course, even after controlling for initial occupational status. This aligns with the intuition that search costs and job attachment increase with experience.

- **Regional Variation:** Mobility patterns vary across regions, consistent with varying local industrial structures and labour market conditions. Areas such as London City and Yorkshire experienced higher mobility, whereas areas like parts of Durham and Leicestershire experienced lower mobility.

- **Cohort Variation:** The 1881-1901 and 1891-1911 cohorts have consistently higher probabilities of upgrade compared to the 1861-81 cohort. This suggests that mobility was more limited in the 1861 cohort, but later cohorts experienced higher levels of flexibility in the market. This may be due to industrial change or advancements in transportation, which reduced the cost of changing jobs or commuting to better jobs outside of one's commuting area.

Across all models, these results suggest that occupational mobility, even during periods of rapid economic change and growth, is relatively limited within cohorts. Factors such as age, location, and initial status play a significant role in determining occupational mobility.

4.1 Identification Strategy and Empirical Specification

The first model estimates how individual characteristics predict labour market transitions over a twenty-year horizon. Specifically, I estimate logistic regressions of the form:

$$\Pr(\text{Upgrade}_{ir,t+1}) = \beta_0 + \beta_1 \cdot \text{CensusYear}_t + \beta_2 \cdot \mathbf{X}'_{i,t} + \varepsilon_{i,t} \tag{5}$$

Where:

- $\Pr(\text{Upgrade}_{i,t+1})$ is the probability of occupational upgrade for individual i in the 20-year later sample, $t+1$.

- $CensusYear_t$ captures variation in mobility across censuses in start year t. $\mathbf{X}_{i,t}$ is a vector of controls: marital status, unemployment, disability, and household size during start year t.[94]

I then expand this to test whether adjustment varies by age group and region. Younger workers would be expected to have less attachment to their occupation, and therefore would be more willing to incur retraining or reskilling costs. To measure the age and regional effects of labour market change, I estimate the following equation:

$$\Pr(\text{Upgrade}_{ir,t+1}) = \beta_0 + \beta_1 \cdot \text{CensusYear}_t + \beta_2 \cdot \text{AgeGroup}_{i,t} + \beta_3 \cdot \text{Region}_{i,t} + \beta_4 \cdot \mathbf{X}'_{i,t} + \varepsilon_{it} \quad (6)$$

Where:

- $\text{AgeGroup}_{i,t}$ is the individual worker's (i) age group in the initial census year t, (18–25 through 65+).
- $\text{Region}_{i,t}$ indicates the individual worker's (i) registration county and district in start year t.

To measure conditional mobility, I adjust Equations (5) and (6) by adding in the factor variable $\text{HISCLASS}_{i,t}$, which captures a worker's initial occupational class. Equation (5) therefore becomes:

$$\Pr(\text{Upgrade}_{ir,t}) = \beta_0 + \beta_1 \cdot \text{CensusYear}_t + \beta_2 \cdot \text{HISCLASS}_{i,t} + \beta_4 \cdot \mathbf{X}'_{i,t} + \varepsilon_{it} \quad (7)$$

And Equation (6) becomes:

$$\Pr(\text{Upgrade}_{irt}) = \beta_0 + \beta_1 \cdot \text{CensusYear}_t + \beta_2 \cdot \text{HISCLASS}_{i,t} + \beta_3 \cdot \text{AgeGroup}_{i,t} + \beta_4 \cdot \text{Region}_{i,t} + \beta_5 \, \mathbf{X}'_{i,t} + \varepsilon_{irt} \quad (8)$$

All models are weighted by inverse propensity score weights to correct for linking biases, as described in Section 3. These models seek to disentangle how various factors, namely

94. Unless otherwise noted, these controls are used throughout. Unemployment status combines *inactive* codes and occupation strings; for instance, students were flagged as not working but not seeking work. Schurer et al., *Integrated Census Microdata (I-CeM), 1851-1911*; Schurer and Higgs, *Integrated Census Microdata (I-CeM) Names and Addresses, 1851-1911*

region and age, contribute to occupational mobility and employment in England and Wales from 1861 to 1911.

4.2 Unconditional Mobility

I first examine the determinants of unconditional mobility, focusing on what generally influenced occupational upgrading across cohorts. I begin by comparing cohorts to see which generations experienced higher probabilities of upgrading over time, then consider life-cycle and regional factors. This period coincides with the continuing, large-scale shift of the British labour force out of agriculture and into factory-based industries that began in the First Industrial Revolution. Such structural change should, in principle, have created scope for upward occupational mobility, as agricultural jobs were generally less mobile than unskilled factory work.[95] As I show below, the extent of upgrading was neither uniform across regions nor equally distributed across the life cycle.

Table 5 reports baseline logit models of occupational upgrading for three linked census samples: 1861–81, 1881–1901, and 1891–1911.

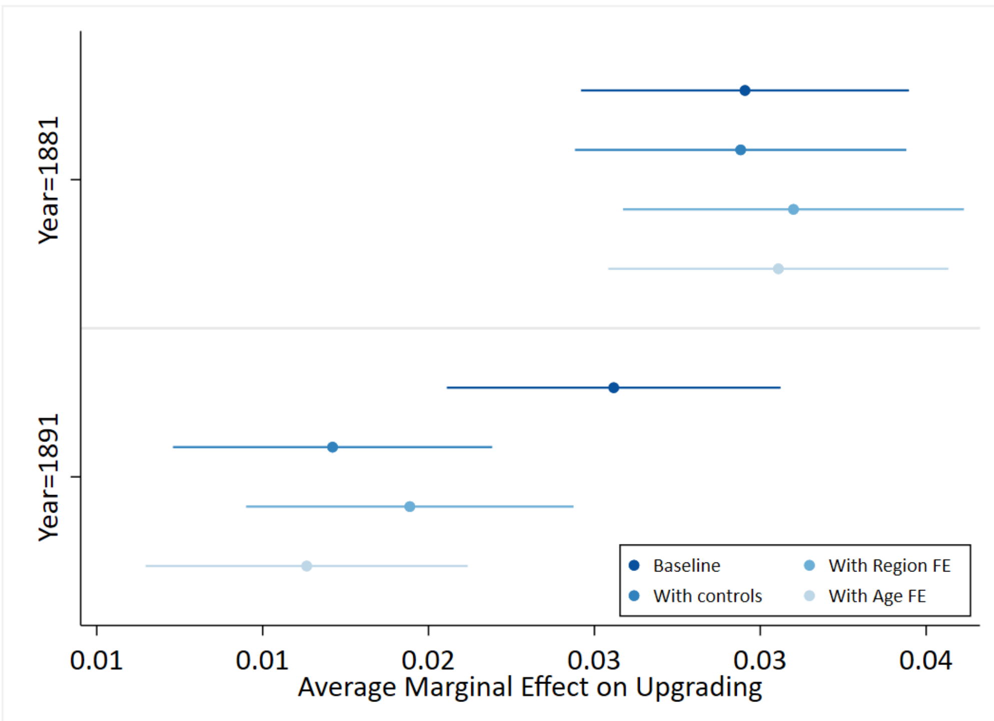
The results in Table 5 indicate that occupational mobility was relatively limited. Although mobility was significantly higher in both the 1881-1901 and 1891-1911 cohorts, Figure 8 shows that the cohort-level marginal effects are small. Adjusting for regional effects in Model (3), the positive relationship between the 1881-1901 and 1891-1911 cohorts and the probability of an upgrade slightly increases. I present average marginal effects by region in Figure 9, demonstrating that access to opportunity varied significantly by geography.

Adjusting these estimates for age effects further demonstrates that occupational mobility was not homogeneous for all workers. As shown in Model (4) of Table 5, older cohorts had lower probabilities of upgrade compared to workers aged 18-25. These results suggest that individuals are less likely to experience upward mobility over the life cycle, despite gaining experience in the market.

Together, these regressions suggest two key points. First, occupational mobility was relatively low across England and Wales from 1861 to 1911. Second, factors such as age and region play a significant role in predicting mobility. Persistent age and regional differences

95. Collins and Zimran, "Working Their Way Up?," pp. 240-41

Figure 8: Unconditional Mobility: Average Marginal Effect by Census Year



Notes: Table 5, Schurer et al., *Integrated Census Microdata (I-CeM), 1851-1911*; Schurer and Higgs, *Integrated Census Microdata (I-CeM) Names and Addresses, 1851-1911*; Marco HD Van Leeuwen and Maas, *HISCLASS: A historical international social class scheme*; Lambert et al., "The Construction of HISCAM," pp. 77-89

Table 5: Unconditional Mobility, 1861-1911

	(1) Pr(Upgrade)	(2) Pr(Upgrade)	(3) Pr(Upgrade)	(4) Pr(Upgrade)
1861-1881 Sample	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)
1881-1901 Sample	0.142***	0.143***	0.152***	0.150***
	(0.01)	(0.01)	(0.01)	(0.01)
1891-1911 Sample	0.124***	0.084***	0.096***	0.081***
	(0.01)	(0.01)	(0.01)	(0.01)
Age 18-25				0.000
				(.)
Age 26-35				-0.161***
				(0.01)
Age 36-45				-0.264***
				(0.01)
Age 46-55				-0.364***
				(0.01)
Age 56-65				-0.383***
				(0.07)
Constant	-0.950***	-0.802***	-0.565***	-0.488***
	(0.01)	(0.01)	(0.01)	(0.01)
Controls	No	Yes	Yes	Yes
Region FE	No	No	Yes	Yes
Observations	3,434,046	3,374,021	3,374,021	3,374,021
AIC	3,064,379	2,970,701	2,961,571	2,956,600

Notes: Variation in sample size come from coverage in HISCLASS, HISCAM-GB, and demographic controls. Schurer et al., *Integrated Census Microdata (I-CeM), 1851-1911*; Schurer and Higgs, *Integrated Census Microdata (I-CeM) Names and Addresses, 1851-1911*; Marco HD Van Leeuwen and Maas, *HISCLASS: A historical international social class scheme*; Lambert et al., "The Construction of HISCAM," pp. 77-89

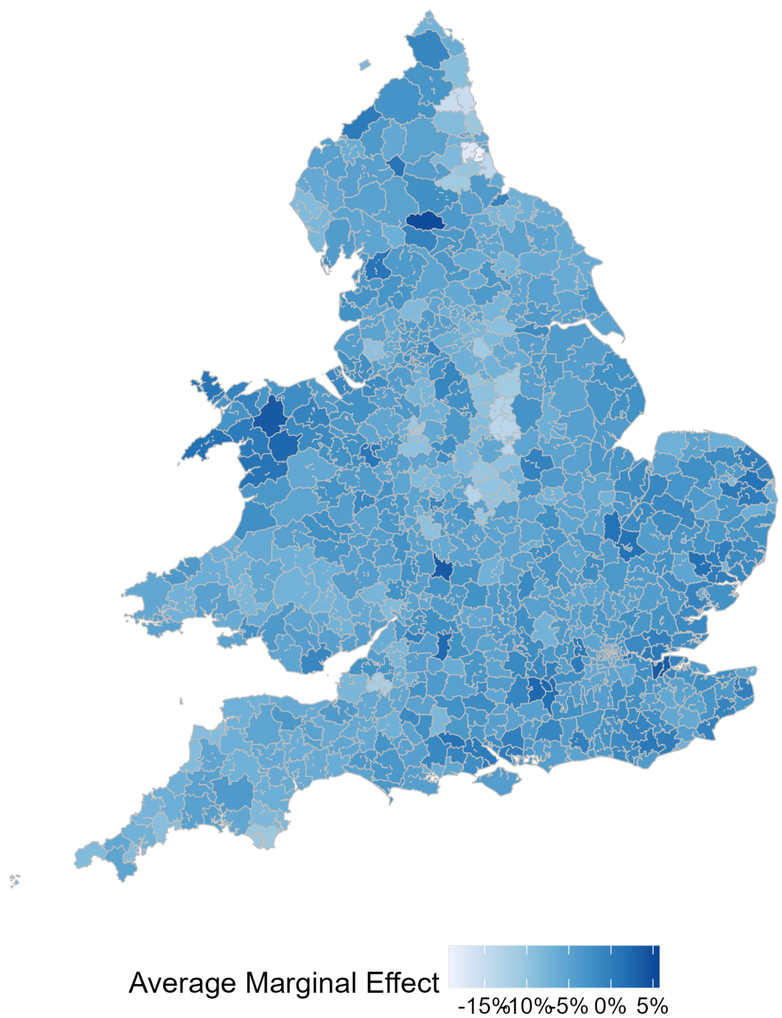
point to the importance of local economic structures in shaping workers' opportunities, especially in settings where migration is limited.

4.3 Conditional Mobility

Unconditional mobility may obscure class differences in upgrading. For example, a large proportion of high-status workers in a given area may obscure effects for mid-status workers.

To adjust for conditional mobility, I recalculate regressions from above with an additional HISCLASS measure. The simple model is presented in Table 6, and age effects are presented

Figure 9: Unconditional Mobility: Average Marginal Effect by Region



Notes: Table 5, Model (4). Schurer et al., *Integrated Census Microdata (I-CeM), 1851-1911*; Schurer and Higgs, *Integrated Census Microdata (I-CeM) Names and Addresses, 1851-1911*; Marco HD Van Leeuwen and Maas, *HISCLASS: A historical international social class scheme*; Lambert et al., "The Construction of HISCAM," pp. 77-89. GIS boundary files were provided by Max Satchell and were constructed from parish files by Joe Day for the ESRC fertility project directed by Alice Reid. Satchell et al., *1851 England and Wales census parishes, townships and places*

in Table 7. Figure 10 confirms conventional wisdom—upgrading among higher white collar workers and farmers is relatively low, whereas unskilled and semiskilled experience the largest boost to mobility.[96] These effects are robust to demographic and regional controls.

After controlling for initial class, age effects are smaller, but do not disappear for cohorts aged 26-55. These results suggest that mobility is still lower for older individuals, even after

96. Collins and Zimran, "Working Their Way Up?," pp. 240-41

Table 6: Conditional Mobility, 1861-1911

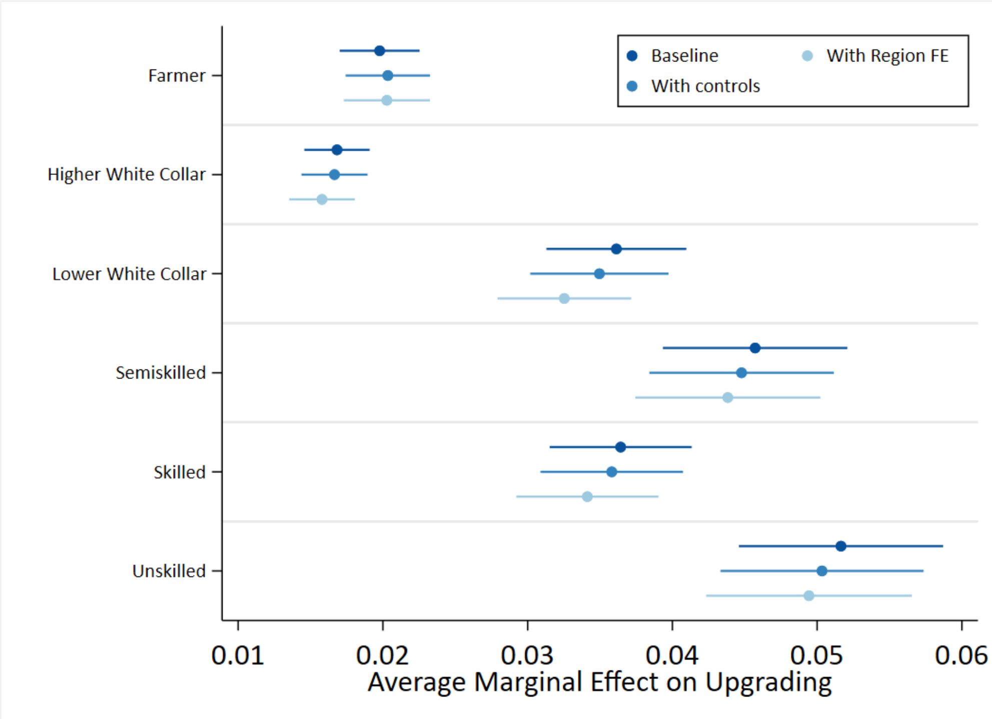
	(1) Pr(Upgrade)	(2) Pr(Upgrade)	(3) Pr(Upgrade)
1861-1881 Sample	0.000	0.000	0.000
	(.)	(.)	(.)
1881-1901 Sample	0.216***	0.211***	0.207***
	(0.02)	(0.02)	(0.02)
1891-1911 Sample	0.198***	0.154***	0.145***
	(0.02)	(0.02)	(0.02)
Farmer	0.000	0.000	0.000
	(.)	(.)	(.)
Higher White Collar	-0.198***	-0.249***	-0.313***
	(0.03)	(0.03)	(0.03)
Lower White Collar	0.865***	0.788***	0.687***
	(0.02)	(0.02)	(0.03)
Semiskilled	1.348***	1.301***	1.300***
	(0.03)	(0.03)	(0.03)
Skilled	0.880***	0.831***	0.770***
	(0.02)	(0.02)	(0.03)
Unskilled	1.750***	1.698***	1.742***
	(0.03)	(0.03)	(0.03)
Constant	-2.288***	-2.080***	-1.774***
	(0.02)	(0.02)	(0.02)
Controls	No	Yes	Yes
Region FE	No	No	Yes
Observations	2,800,894	2,791,598	2,791,598
AIC	2,354,872	2,338,607	2,324,405

Notes: Variation in sample size comes from coverage in HISCLASS, HISCAM-GB, and demographic controls. Schurer et al., *Integrated Census Microdata (I-CeM), 1851-1911*; Schurer and Higgs, *Integrated Census Microdata (I-CeM) Names and Addresses, 1851-1911*; Lambert et al., "The Construction of HISCAM," pp. 77-89; Marco HD Van Leeuwen and Maas, *HISCLASS: A historical international social class scheme*

controlling for initial class. As such, lower mobility throughout the life cycle cannot be explained by high rates of occupational upgrading in entry-level jobs alone.

All together, these results suggest that age, region, and initial class play a significant role in determining access to opportunity. While older workers have more experience, my results imply that mobility decreases throughout the life cycle, even when accounting for initial class. Further, mobility varies significantly by region, highlighting uneven access to rapidly upgrading jobs during the Second Industrial Revolution. As such, occupational mobility varies significantly by demographic and geographic characteristics.

Figure 10: Conditional Mobility: Average Marginal Effect by HISCLASS



Notes: Table 6, Schurer et al., *Integrated Census Microdata (I-CeM), 1851-1911*; Schurer and Higgs, *Integrated Census Microdata (I-CeM) Names and Addresses, 1851-1911*; Marco HD Van Leeuwen and Maas, *HISCLASS: A historical international social class scheme*; Lambert et al., "The Construction of HISCAM," pp. 77-89

5 Post-Shock Determinants of Occupational Upgrading

This section examines how temporary displacement during the Cotton Famine (1861–1865) shaped workers' long-run labour market outcomes. In this section, I find the following:

- **Cohort-Level Employment Effects:** When examining cohort-level data, the Cotton Famine has a small but significantly positive impact on cotton employment from 1861 to 1881. This suggests that the trade shock did not permanently lower cotton employment.
- **Overall Mobility Effects:** Limiting analysis to the linked sample, the Cotton Famine is related to an increase in mobility, even after adjusting for demographic controls and regional variation. This suggests that workers were able to recover from the brief shock to cotton in aggregate.
- **Mobility for Cotton Workers in Cotton Towns:** However, individuals employed in cotton in 1861 in areas with higher levels of pre-shock cotton employment experienced significantly lower mobility than other workers. This suggests that workers in industry

Table 7: Conditional Mobility: Age Effects, 1861-1911

	(1) Pr(Upgrade)
1861-1881 Sample	0.000
	(.)
1881-1901 Sample	0.205***
	(0.02)
1891-1911 Sample	0.136***
	(0.02)
Farmer	0.000
	(.)
Higher White Collar	-0.333***
	(0.03)
Lower White Collar	0.651***
	(0.03)
Semiskilled	1.255***
	(0.03)
Skilled	0.726***
	(0.03)
Unskilled	1.703***
	(0.03)
Age 18-25	0.000
	(.)
Age 26-35	-0.096***
	(0.01)
Age 36-45	-0.176***
	(0.01)
Age 46-55	-0.280***
	(0.02)
Age 56-65	-0.111
	(0.08)
Constant	-1.690***
	(0.02)
Controls	Yes
Region FE	Yes
Observations	2,791,598
AIC	2,322,273

Notes: Variation in sample size comes from coverage in HISCLASS, HISCAM-GB, and demographic controls. Schurer et al., *Integrated Census Microdata (I-CeM), 1851-1911*; Schurer and Higgs, *Integrated Census Microdata (I-CeM) Names and Addresses, 1851-1911*; Lambert et al., "The Construction of HISCAM," pp. 77-89; Marco HD Van Leeuwen and Maas, *HISCLASS: A historical international social class scheme*

towns faced lasting mobility impacts despite the brief trade shock.

These findings indicate that, while unaffected workers recovered, workers displaced by trade shocks experienced effects that persisted for 20 years, showing that even brief displacement can have long-term consequences.

5.1 Identification Strategy and Empirical Specification

Section 4 estimates the general determinants of occupational upgrading from 1861 to 1911, but does not explain how workers respond to specific episodes of economic change. As such, I expand my analysis to focus on mobility after one of these shocks—the Cotton Famine.

First, I estimate district-level regressions of changes in cotton employment shares on the Cotton Shock.[97] The key exposure variable is defined above in Equation (4). Subsection 5.2 presents these results using the first-differences specification:

$$\Delta\text{CottonShare}_{r,1861-81} = \alpha + \beta_1 \cdot \text{CottonShock}^{UK}_{r,c,1861-65} + \epsilon \tag{9}$$

This model assesses whether the shock affected long-term district-level employment in both the cohort-level and linked samples.

I then estimate my primary model of the shock's impact on individual occupational mobility. Equation 10 is a shift-share-style regression, relating labour-share-weighted trade shocks to mobility, a common approach in labour economics to exogenously weight shocks by the share of workers exposed.[98] Whereas Arthi et al. use a binary "Cotton Town" indicator to measure the shock, my approach allows exposure to scale with the share of workers employed in cotton.[99] The identifying assumption is that regions more specialised in cotton were more exposed to the Northern blockade, while less cotton-heavy areas were relatively insulated. This pre-shock variation captures regional impacts of the trade disruption.

I then estimate the effect of this exposure on individual outcomes as follows:

$$\Pr(\text{Upgrade}_{ir,1881}) = \alpha + \beta_1 \cdot \text{CottonShock}^{UK}_{r,c,1861-65} + \beta_2 \cdot \mathbf{X}'_{ir,1861} + \delta_{r,1861} + \varepsilon_{ir} \tag{10}$$

Where:

- $\Pr(\text{Upgrade}_{ir,1881})$ is the probability of occupational upgrade in 1881 for individual i in

97. Here, $\Delta\text{CottonShare}_{r,1861-81} = log(\frac{L_{r,c,1881}}{L_{r,1881}}) - log(\frac{L_{r,c,1861}}{L_{r,1861}})$.

98. Autor, Dorn, and Hanson, "The China Syndrome," 2121-2168; Kirill Borusyak, Peter Hull, and Xavier Jaravel, "A Practical Guide to Shift-Share Instruments," *National Bureau of Economic Research Working Paper Series* No. 33236 (2024), doi:10.3386/w33236; David Card, "Immigration and Inequality," *American Economic Review* 99, no. 2 (April 2009): 1–21, ISSN: 0002-8282, doi:10.1257/aer.99.2.1

99. I include robustness checks to Arthi et al.'s shock variable in Appendix A. Arthi, Beach, and Hanlon, "Recessions, Mortality, and Migration Bias," p. 238

region r.

- $\text{CottonShock}^{UK}_{r,c,1861-65}$ measures local exposure to the Cotton Famine.
- $\mathbf{X}_{ir,1861}$ is a vector of controls: marital status, unemployment, disability, and household size.
- $\delta_{r,1861}$ are region-level fixed effects.

All regressions on the linked sample are weighted by inverse propensity score weights to correct for linking biases, as described in Section 3.

5.2 Changes in the Share of Cotton Employment

Examining mobility across England and Wales may obscure the occupation-level shifts occurring as new technology creates more jobs while eliminating others. Therefore, I present results on occupational upgrading post-shock.

First, I aggregate cohort-level samples by registration districts to estimate the impact of the Cotton Famine on the share of cotton employment across England and Wales. These "Cohort Samples" encompass all adult male census data points, not simply those that were linked. I summarise this estimation in Equation (11).

$$\Delta\text{CottonShare}_r = \beta_0 + \beta_1 \cdot \text{CottonShock}^{UK}_{r,c,1861-65} + \boldsymbol{\gamma}'\mathbf{X}_r + \varepsilon_r \tag{11}$$

where:

- $\Delta\text{CottonShare}_r$ is the log change in the share of the adult population employed in cotton in region r,
- $\text{CottonShock}^{UK}_{r,c,1861-65}$ captures variation in local exposure to the Cotton Famine.
- $\mathbf{X}_r$ is a vector of pre-shock characteristics, including shares of other industries (wool, silk, linen, other textiles) and demographic/economic indicators (share disabled and share unemployed),
- $\boldsymbol{\gamma}$ is the vector of coefficients on the controls,
- ε_r is the error term.

The results of this estimation are shown in Table 8, and are weighted by registration-district-level population.

Table 8: Impact of Cotton Shock on Cotton Employment by Sample

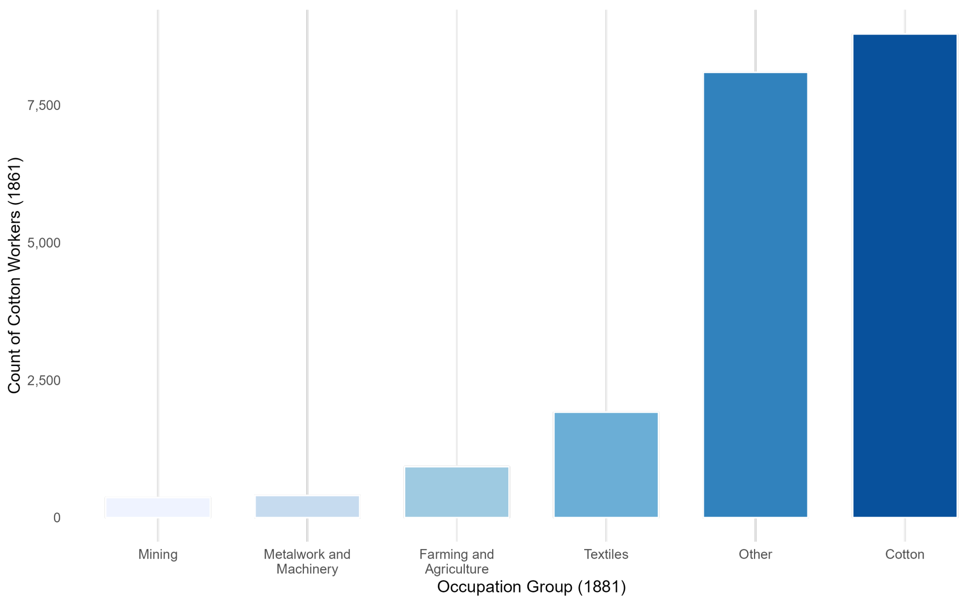
	Cohort Sample (1)	Cohort Sample (2)	Linked Sample (3)	Linked Sample (4)
Cotton Shock (1861-65)	0.006^{**}	0.008^{***}	-0.478	-0.297
	(0.002)	(0.003)	(0.323)	(0.341)
Constant	-0.582^{***}	-0.737^{***}	-0.386^{**}	-0.619^{**}
	(0.085)	(0.267)	(0.157)	(0.248)
Controls	No	Yes	No	Yes
Observations	597	597	597	597

Notes: K. Schurer et al., *Integrated Census Microdata (I-CeM), 1851-1911*, 2025, doi:10.5255/UKDA-SN-7481-3; K. Schurer and E. Higgs, *Integrated Census Microdata (I-CeM) Names and Addresses, 1851-1911: Special Licence Access*, 2025, doi:10.5255/UKDA-SN-7856-2; Marco HD Van Leeuwen and Maas, *HISCLASS: A historical international social class scheme*, Mitchell and Deane, *Abstract of British Historical Statistics*, p.180-81

The results in Table 8 suggest that, without limiting the sample to linked individuals, regions with a higher exposure to the cotton trade shock experienced relatively small increases in their share of cotton employment in 1881. While these coefficients are significant, they are very tiny—Model (2) suggests a 1-unit increase in the shock is only correlated with a 0.8 percentage point increase in the share of workers employed in cotton, controlling for the share of workers who are disabled and/or unemployed. This result is not altogether shocking—the Cotton Famine was brief, and cotton imports recovered briefly after the American Civil War, as shown in Figure 6. However, these regressions are limited in capturing the actual effects on the workforce at the time of the shock. Because the dependent variable measures changes in regional cotton share rather than outcomes for individual workers, it cannot isolate what happened to the cohort of workers employed in cotton before the shock. For instance, regions could see a rise in cotton production driven by new entrants or migration, which would obscure effects on the original cohort while boosting overall employment shares. Consequently, while the coefficients on decomposed shocks provide suggestive evidence of regional adjustments, they do not directly measure labour-market outcomes for the pre-shock cotton workforce, highlighting an inherent limitation of using cohort data.

To control for possible biases from migration and new entrants, I rerun the regressions from Table 8 on the linked sample. In Models (3) and (4), I rerun the share of cotton employment on the cotton shock, and find insignificant results. The insignificant coefficients on the cotton shock suggest that the trade shock had a null effect on cotton employment in the sample. This result suggests that the brief interruption in cotton supply from 1861 to 1865 did not affect what percentage of workers were employed in cotton. Put another way, men of adult working age were not, in the long run, pulled away from cotton after the shock. These results suggest that the region-level composition of the workforce was not heavily impacted by the 1861-65 Cotton Famine. This is consistent with the data presented in Figure 11, which shows that many cotton workers in 1861 remained working in cotton in 1881.

Figure 11: Count of Cotton Workers (1861) by Occupational Groups (1881)



Notes: Schurer et al., *Integrated Census Microdata (I-CeM), 1851-1911*, Schurer and Higgs, *Integrated Census Microdata (I-CeM) Names and Addresses, 1851-1911*; Marco HD Van Leeuwen and Maas, *HISCLASS: A historical international social class scheme*, Lambert et al., "The Construction of HISCAM," pp. 77-89; Mitchell and Deane, *Abstract of British Historical Statistics*, p.180-81

Together, these results suggest that the Cotton Famine had little to no effect on the share of workers employed in cotton twenty years later. This may be because workers were able to move into cotton once the industry recovered, suggesting the trade shock did not leave workers with a distaste for cotton compared to other industries.

5.3 Occupational Upgrading and the Cotton Shock

While the results in Subsection 5.2 suggest the Cotton Famine did not impact employment in cotton in 1881, it does not provide evidence on the impact of the Cotton Famine on individual employment outcomes. To test whether individual workers who were working during the Cotton Famine in affected regions experienced significant mobility effects 20 years later, I measure the impact of the shock on the probability of occupational upgrading in Table 9.

Table 9: Impact of Cotton Shock on Occupational Upgrading

	(1) Pr(Upgrade)	(2) Pr(Upgrade)
Cotton Shock (1861-65)	0.527***	0.574***
	(0.07)	(0.07)
Constant	-0.977***	-0.883***
	(0.01)	(0.01)
Controls	No	Yes
Observations	663,951	640,246
AIC	651,015	619,095

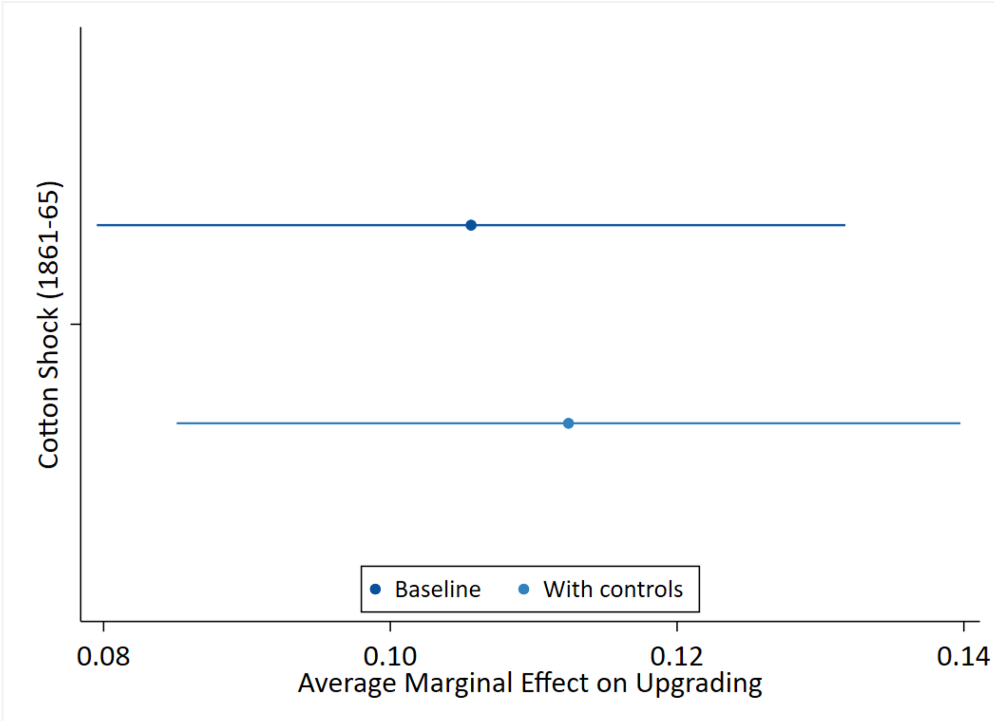
Notes: Variation in sample size come from coverage in HISCAM-GB and demographic controls. Schurer et al., *Integrated Census Microdata (I-CeM), 1851-1911*; Schurer and Higgs, *Integrated Census Microdata (I-CeM) Names and Addresses, 1851-1911*; Lambert et al., "The Construction of HISCAM," pp. 77-89; Marco HD Van Leeuwen and Maas, *HISCLASS: A historical international social class scheme*, Mitchell and Deane, *Abstract of British Historical Statistics*, p.180-81

Table 9 measures the impact of the Cotton Shock, weighted by the initial share of cotton employment, on workers' probability of occupational upgrade. Model (2) controls for various demographic differences, including household size, unemployment, marital status, and disability. These results suggest that mobility was relatively low in the 1861-81 cohort, consistent with Table 5, but was in fact higher in areas hit harder by the Cotton Famine. The estimated marginal effect implies that doubling exposure to the cotton trade shock (e.g., a one-unit increase in the log shock) raises the odds of upgrade by approximately 69 per cent, or the predicted probability of upgrade to around 12 percentage points.[100]

This pattern may reflect growth in non-cotton industries in towns most affected by the cotton trade shock, rather than a direct benefit to cotton workers, who, as shown in Table

100. *See* Figure 12.

Figure 12: Average Marginal Effect of Cotton Shock



Notes: Table 9, Schurer et al., *Integrated Census Microdata (I-CeM), 1851-1911*; Schurer and Higgs, *Integrated Census Microdata (I-CeM) Names and Addresses, 1851-1911*; Lambert et al., "The Construction of HISCAM," pp. 77-89; Marco HD Van Leeuwen and Maas, *HISCLASS: A historical international social class scheme*, Mitchell and Deane, *Abstract of British Historical Statistics*, p.180-81

10, experienced little or no increase in mobility. For instance, towns with high exposure to the cotton trade shock may have had the industrial infrastructure to recover quickly as cotton rebounded. As cotton imports fell temporarily, manufacturers and entrepreneurs in these towns could have shifted resources and workers into manufacturing for other goods, such as wool, silk, linen, or emerging consumer goods industries. Rising demand for factory-produced textiles and other consumer goods in the late nineteenth century may have created new employment opportunities, facilitating upward occupational mobility for workers.

Nevertheless, these models do not fully isolate the shock's long-term impact on the occupational mobility of cotton workers in 1861. To measure the impact of the shock on cotton workers only, I regress the probability of occupation upgrading on the interaction between the shock variable and a binary representing whether or not the individual worker was employed in cotton in 1861 ($\text{Cotton}_{i,1861}$). These results are presented in Table 10.

Table 10 presents four models: Model (1) simply presents the results from Table 9 with

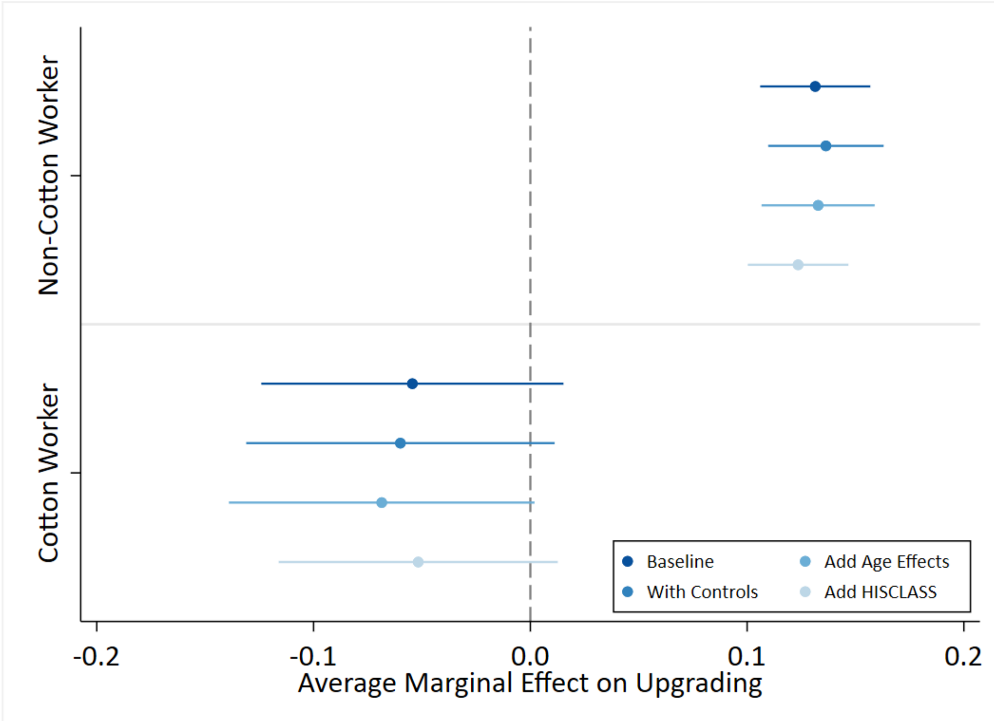
Table 10: Impact of Cotton Shock on Occupational Upgrading by Cotton Worker Status

	(1) Pr(Upgrade)	(2) Pr(Upgrade)	(3) Pr(Upgrade)	(4) Pr(Upgrade)
Cotton Shock (1861-65)	0.656***	0.697***	0.680***	0.663***
	(0.06)	(0.07)	(0.07)	(0.06)
Non-Cotton Worker	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)
Cotton Worker	0.292***	0.338***	0.334***	0.109
	(0.07)	(0.07)	(0.07)	(0.07)
Non-Cotton Worker × Cotton Shock (1861-65)	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)
Cotton Worker × Cotton Shock (1861-65)	-0.901***	-0.968***	-0.991***	-0.931***
	(0.17)	(0.17)	(0.17)	(0.18)
Age 18–25			0.000	0.000
			(.)	(.)
Age 25–35			-0.099***	-0.044***
			(0.01)	(0.01)
Age 35–45			-0.225***	-0.140***
			(0.01)	(0.01)
Age 45–55			-0.374***	-0.281***
			(0.02)	(0.02)
Age 55–65			-0.303***	-0.187*
			(0.07)	(0.08)
Farmer				0.000
				(.)
Higher White Collar				-0.007
				(0.04)
Lower White Collar				0.984***
				(0.03)
Semiskilled				1.427***
				(0.03)
Skilled				0.886***
				(0.03)
Unskilled				1.261***
				(0.04)
Constant	-0.980***	-0.886***	-0.849***	-2.073***
	(0.01)	(0.01)	(0.01)	(0.03)
Controls	No	Yes	Yes	Yes
Observations	663,951	640,246	640,246	545,426
AIC	650,926	618,998	617,741	504,815

Notes: Variation in sample size come from coverage in HISCAM-GB and demographic controls. K. Schurer et al., *Integrated Census Microdata (I-CeM), 1851-1911*, 2025, doi:10.5255/UKDA-SN-7481-3; K. Schurer and E. Higgs, *Integrated Census Microdata (I-CeM) Names and Addresses, 1851-1911: Special Licence Access*, 2025, doi:10.5255/UKDA-SN-7856-2; Lambert et al., "The Construction of HISCAM," pp. 77-89; Marco HD Van Leeuwen and Maas, *HISCLASS: A historical international social class scheme*, Mitchell and Deane, *Abstract of British Historical Statistics*, p.180-81

the additional interaction term between $\text{Cotton}_{i,1861}$ and the shock exposure variable. Model (2) adds the same demographic controls to Model (1), as previously defined. Model (3) adds

Figure 13: Average Marginal Effect of Cotton Shock by Cotton Worker Status



Notes: Table 10, Schurer and Higgs, *Integrated Census Microdata (I-CeM) Names and Addresses, 1851-1911*; Lambert et al., "The Construction of HISCAM," pp. 77-89; Marco HD Van Leeuwen and Maas, *HISCLASS: A historical international social class scheme*, Mitchell and Deane, *Abstract of British Historical Statistics*, p.180-81

age group covariates, which control for life-cycle upgrading effects. Model (4), finally, controls for variation across HISCLASS groups.

I plot the average marginal effects of the trade shock by whether or not the worker was employed in cotton in 1861 in Figure 13. Two things are clear from these results: first, cotton workers experienced significantly lower rates of mobility than non-cotton workers in areas hit harder by the trade shocks. Second, on average, the shock reduces the probability of upgrading into a better occupation among cotton workers. These results suggest that cotton workers did not benefit from any boost correlated with areas hit hardest by the Cotton Famine.

The Cotton Famine, in aggregate, did not have lasting impacts on the share of workers employed in cotton, nor did it greatly inhibit occupational mobility in regions most exposed to the shock. However, cotton workers in areas more exposed to the trade shock were more experienced lower rates of occupational mobility. These results suggest that individuals working in affected industries during the shock retain scars of the shock, despite the rest of the cohort rebounding.

6 Discussion

Without adequate economic opportunity across the life cycle, workers remain in jobs that poorly match their skills, leading to underdeveloped human capital and allocation inefficiencies. These inefficiencies inhibit economic growth, as individuals are unable to respond efficiently to changing incentives across emerging industries. Beyond economics, mobility matters for social equity and cohesion—if one's economic outcome is determined solely by parental status or geography, workers have little incentive to invest in their skills or create further opportunity. While economic mobility is historically limited, my results suggest that age, region, and industrial change matter for long-term mobility. In the sections below, I discuss possible mechanisms of mobility and areas for future research.

6.1 Mechanisms of Mobility

The results presented in Section 4 indicate that regional variation played a central role in shaping access to opportunity. In a period when transportation was far more limited than today, access to better employment often hinged on a worker's ability to migrate to areas of rapid industrial growth. While expanding industrial production provided incentives to move, workers faced substantial constraints: poor infrastructure, information asymmetries, and the financial cost of moving. Further, local policy may have provided incentives against moving to opportunity. For example, relatively generous poor relief in regions such as Lancashire may have discouraged out-migration, even during periods of mass unemployment.[101] Together, these findings show that location matters—regional variation strongly affects a worker's occupational mobility.

I also find that age significantly impacts mobility. Earnings generally rise with age, as older workers have higher experience and more developed skills. At the same time, however,

101. Hall, "A Poor Cotton Weyver: Poverty and the Cotton Famine in Clitheroe," pp. 227-250, George Boyer, "Poor Relief, Informal Assistance, and Short Time during the Lancashire Cotton Famine," *Explorations in Economic History* 34, no. 1 (January 1997): 56–76, ISSN: 0014-4983, doi:10.1006/exeh.1996.0663

age increases the costs of job changes, such as reskilling, moving, and adapting to new work. My results suggest that older workers, even those aged 25 to 35, experience lower mobility. While it is possible this is due to the increased cost of job change in Victorian England, I argue it is more likely due to decreased returns to skill with age. While clerical and white collar jobs existed in Victorian England, much work still relied on physical labour.[102] Declining mobility with age may, then, be related to industry valuing physical strength over cognitive skill. These results suggest that returns to experience in my sample were limited or did not increase significantly with age.

6.2 Mobility after an Adverse Shock

Section 5 extends this analysis by examining how workers responded to a localised trade shock in their industry. Out of this analysis, two main findings emerge: the trade shock did not adversely impact employment or mobility in affected areas, but workers employed in the affected industry at the time of the shock saw decreased mobility.

The first result is consistent with the short-lived and geographically concentrated nature of the Cotton Famine. Increased mobility in areas affected by the shock is surprising, but not unfathomable. Areas impacted by the cotton shock were also industrial centres on finance and trade—it is possible that these professions continued to grow and create further employment opportunities for low- and mid-skill workers long after the Cotton Famine. Taken together, these results suggest that areas hit hardest by the Cotton Famine were able to recover within the 20-year window observed.

However, this analysis does not capture what happened to those employed in cotton. Though temporary displacement may not have been a significant issue for non-cotton workers, the results in Table 10 suggest that cotton workers in large cotton towns experienced lasting hits to their mobility. In fact, Figure 13 suggests that these workers experienced little to no upward mobility 20 years later. This suggests that adverse shocks, even brief episodes, can have longer-term impacts on worker outcomes.

There are several reasons cotton workers could not recover. First, it is possible that factories shifted production in ways that did not align with the existing set of cotton workers'

102. Berg, *The Age of Manufactures, 1700-1820*, p. 219

skills. If cotton workers' skills, for example, were directly tied to production methods with American cotton, shifts to non-American cotton could be biased against their skillset. Moreover, workers in those regions may have migrated away from industrial centres to look for work. Since these workers were no longer in the centre of the cotton industry, they would not benefit from the external economies of scale that made Lancashire cotton more productive. While my results do not pinpoint the source of limited mobility among cotton workers, they illustrate a key finding: recovery at the regional level does not necessarily imply that affected workers recovered.

These results underscore a central finding of this paper—mobility varies by region, age, and industry. Examining nationwide, or even county-wide, trends obscures the conditions faced by workers. This research contributes to a larger literature examining how heterogeneous labour market experiences impact mobility.

6.3 Further Areas of Research

While this analysis offers new insight into labour market dynamics using a large sample of linked individual outcomes, it also opens avenues for future research. Two particularly promising areas concern gender and migration.

First, this study focuses on adult male workers. This choice reflects both data-linkage challenges and the centrality of men's wages to household income in nineteenth-century Britain, when the male breadwinner model had become dominant.[103] Previous scholarship has noted persistent difficulties in linking women across censuses, particularly due to surname changes at marriage.[104] Nonetheless, women were integral to the textile sector, comprising more than half of the cotton workforce.[105] Extending linkage methods to incorporate female workers would therefore provide a fuller picture of individual labour market trajectories and

103. Sara Horrell and Jane Humphries, "The Origins and Expansion of the Male Breadwinner Family: The Case of Nineteenth-Century Britain," Publisher: Cambridge University Press, *International Review of Social History* 42 (1997): p. 31, ISSN: 00208590, 1469512X

104. James Feigenbaum and Daniel P Gross, "Answering the Call of Automation: How the Labor Market Adjusted to Mechanizing Telephone Operation*," *The Quarterly Journal of Economics* 139, no. 3 (February 2024): 1879–1939, ISSN: 0033-5533, doi:10.1093/qje/qjae005; Emma Diduch, "The Record Linking Glass Ceiling. Applying Automated Methods to the Census and Women's Marriage Records, 1881–1911," *Historical Life Course Studies* 14 (November 2024): 126–143, ISSN: 2352-6343, doi:10.51964/hlcs19189

105. Blaug, "The Productivity of Capital in the Lancashire Cotton Industry during the Nineteenth Century," p. 368

household dynamics.

Second, the present analysis does not examine the role of regional migration in predicting mobility. By combining occupational outcomes with migration patterns, further research could shed light on whether workers migrated in response to labour market change. This analysis could provide insight into the role of migration in shaping worker mobility.

7 Conclusion

As modern economies adjust to rapid technological change and globalisation, understanding the mechanisms underlying mobility has grown increasingly important. Although recent scholarship on labour market outcomes during periods of significant economic change highlights the need to examine these trends across heterogeneous groups, these studies do not capture mobility over the life course. Whereas previous studies have primarily focused on long-term intergenerational mobility, this dissertation leverages a novel linked sample of over four million adult male records across England and Wales from 1861 to 1911 to explore how mobility unfolds across the life cycle.

In doing so, I investigate the impacts of regional and life-cycle variation on mobility, finding that access to opportunity varies significantly across regions and age groups, with older workers experiencing much less mobility than younger workers over time. I then expand this analysis through a case study of mobility after an adverse trade shock, namely the 1861-65 Lancashire Cotton Famine. While this trade shock is associated with upgrading over 20-year worker cohorts, workers employed in the affected industry in 1861 experienced little to no mobility. These results suggest that adverse trade shocks impact mobility over the long run, not simply in the years immediately following the shock.

This study underscores the importance of examining mobility not only across generations but also within individual workers' lifetimes. While neoclassical theory might suggest that workers can easily transition to comparable industries, these findings indicate that local shocks, age, and location can significantly constrain access to opportunity.

This study demonstrates that examining mobility across workers' lifetimes, not just between generations, reveals persistent scarring from economic shocks. While new technologies always bring with them economic change and displacement, long-term worker adjustment remains under-examined. Using a novel database of over four million UK workers across 20-year cohorts, I find cotton workers show reduced mobility twenty years post-displacement, despite regional recovery. These findings suggest short-run studies likely underestimate displacement costs. Future research should identify which policy interventions could break persistent mobility constraints and improve worker outcomes after displacement.

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Appendices

A Robustness Checks

While Sections 4 and 5 capture my primary estimates of occupational upgrading and mobility, the models presented are robust to alternative models. In this section, I present three primary alternative adjustments to my model, using alternative measures of mobility, measures of the cotton shock, and regional effects.

A.1 Alternative Measures of Occupational Mobility (ΔHISCAM)

While there is a large precedent in the literature of measuring occupational upgrading using logit models, these results do not provide context on the scale of upgrades.

As a robustness check, I include alternative regressions for Tables 5 and 6 using the change in HISCAM scores across the two censuses.[106] Instead of a logit model, as presented above, these models present OLS estimates of $\Delta\text{HISCAM}_{ir,t+1}$ on the same factors presented Equations (5), (6), (7), and (8).

Though largely in line with the results I find in Section 4, these alternative models highlight that upgrades were, on average, quite small. However, upgrade effects in Table 12 shows higher-status white collar jobs have a strong negative effect on status 20 years later.

A.2 Alternative Measures of Cotton Shock

In my primary analysis, I use a decomposition of the cotton trade shock, which is sensitive to the reallocation of British cotton imports to non-US sources. However, there is precedent in the literature to instead treat the shock as a binary, defined as a flag whether the town had $> 10\%$ cotton manufacturing employment before the shock.[107]

As a robustness check, I rerun my model from Section 5 using this new shock variable: $\text{CottonShock}^{UK}_{r,c,1861-65} = 1$ if the town registration subdistrict has $> 10\%$ cotton textile manufacturing employment in 1860.

106. These estimates use my adjusted HISCAM measures, detailed in Table 16.
107. Arthi, Beach, and Hanlon, "Recessions, Mortality, and Migration Bias," pp. 230-31

Table 11: Robustness Check: Unconditional Mobility, 1861-1911 (ΔHISCAM)

	(1) Δ HISCAM	(2) Δ HISCAM	(3) Δ HISCAM	(4) Δ HISCAM
1861-1881 Sample	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)
1881-1901 Sample	0.151***	0.148***	0.153***	0.150***
	(0.02)	(0.02)	(0.02)	(0.02)
1891-1911 Sample	0.143***	0.099***	0.107***	0.085***
	(0.02)	(0.02)	(0.02)	(0.02)
Age 18-25				0.000
				(.)
Age 26-35				-0.220***
				(0.02)
Age 36-45				-0.385***
				(0.02)
Age 46-55				-0.513***
				(0.02)
Age 56-65				-0.486*
				(0.22)
Constant	0.618***	0.784***	1.029***	1.138***
	(0.02)	(0.03)	(0.02)	(0.02)
Controls	No	Yes	Yes	Yes
Region FE	No	No	Yes	Yes
Observations	3,434,046	3,374,021	3,374,021	3,374,021
AIC	23,243,232	22,863,175	22,858,013	22,856,671

Notes: Variation in sample size come from coverage in HISCAM-GB and demographic controls. Schurer et al., *Integrated Census Microdata (I-CeM), 1851-1911*; Schurer and Higgs, *Integrated Census Microdata (I-CeM) Names and Addresses, 1851-1911*; Marco HD Van Leeuwen and Maas, *HISCLASS: A historical international social class scheme*; Lambert et al., "The Construction of HISCAM," pp. 77-89

I present results below in Tables 13 and 14:

These results retain the same direction and significance expressed in my primary models, though the negative impact on cotton workers in affected areas is slightly smaller.

B Occupational Data

B.1 Occupational Status Construction

Table 12: Robustness Check: Conditional Mobility, 1861-1911 (ΔHISCAM)

	(1) Δ HISCAM	(2) Δ HISCAM	(3) Δ HISCAM
1861-1881 Sample	0.000	0.000	0.000
	(.)	(.)	(.)
1881-1901 Sample	0.379***	0.372***	0.325***
	(0.03)	(0.03)	(0.02)
1891-1911 Sample	0.349***	0.324***	0.265***
	(0.03)	(0.02)	(0.02)
Farmer	0.000	0.000	0.000
	(.)	(.)	(.)
Higher White Collar	-4.037***	-4.077***	-4.198***
	(0.16)	(0.16)	(0.16)
Lower White Collar	-2.323***	-2.395***	-2.612***
	(0.10)	(0.10)	(0.09)
Semiskilled	1.903***	1.875***	1.872***
	(0.05)	(0.05)	(0.05)
Skilled	0.963***	0.938***	0.857***
	(0.05)	(0.05)	(0.05)
Unskilled	2.595***	2.562***	2.671***
	(0.06)	(0.06)	(0.06)
Constant	-0.663***	-0.546***	-0.154***
	(0.03)	(0.03)	(0.04)
Controls	No	Yes	Yes
Region FE	No	No	Yes
Observations	2,800,894	2,791,598	2,791,598
AIC	18,873,809	18,809,861	18,799,712

Notes: Variation in sample size come from coverage in HISCLASS, HISCAM-GB, and demographic controls. Schurer et al., *Integrated Census Microdata (I-CeM), 1851-1911*; Schurer and Higgs, *Integrated Census Microdata (I-CeM) Names and Addresses, 1851-1911*; Marco HD Van Leeuwen and Maas, *HISCLASS: A historical international social class scheme*; Lambert et al., "The Construction of HISCAM," pp. 77-89

Table 13: Robustness Check: Impact of Cotton Shock on Occupational Upgrading, Alternative Shock Measure

	(1) Pr(Upgrade)	(2) Pr(Upgrade)
CottonShock	0.226***	0.252***
	(0.02)	(0.02)
Constant	-0.973***	-0.879***
	(0.01)	(0.01)
Controls	No	Yes
Observations	663,951	640,246
AIC	651,077	619,146

Notes: Schurer et al., *Integrated Census Microdata (I-CeM), 1851-1911*; Schurer and Higgs, *Integrated Census Microdata (I-CeM) Names and Addresses, 1851-1911*; Marco HD Van Leeuwen and Maas, *HISCLASS: A historical international social class scheme*; Lambert et al., "The Construction of HISCAM," pp. 77-89

Table 14: Robustness Check: Impact of Cotton Shock on Occupational Upgrading by Cotton Worker Status, Alternative Shock Measure

	(1) Pr(Upgrade)	(2) Pr(Upgrade)	(3) Pr(Upgrade)	(4) Pr(Upgrade)
CottonShock = 0	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)
CottonShock = 1	0.260***	0.283***	0.276***	0.261***
	(0.02)	(0.02)	(0.02)	(0.03)
Non-Cotton Worker	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)
Cotton Worker	0.334***	0.382***	0.377***	0.163
	(0.09)	(0.09)	(0.09)	(0.09)
CottonShock = 0 × Non-Cotton Worker	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)
CottonShock = 0 × Cotton Worker	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)
CottonShock = 1 × Non-Cotton Worker	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)
CottonShock = 1 × Cotton Worker	-0.434***	-0.472***	-0.484***	-0.464***
	(0.09)	(0.10)	(0.10)	(0.10)
Age 18–25			0.000	0.000
			(.)	(.)
Age 25–35			-0.099***	-0.044***
			(0.01)	(0.01)
Age 35–45			-0.226***	-0.140***
			(0.01)	(0.01)
Age 45–55			-0.374***	-0.281***
			(0.02)	(0.02)
Age 55–65			-0.304***	-0.188*
			(0.07)	(0.08)
Farmer				0.000
				(.)
Higher White Collar				-0.008
				(0.04)
Lower White Collar				0.984***
				(0.03)
Semiskilled				1.429***
				(0.03)
Skilled				0.886***
				(0.03)
Unskilled				1.257***
				(0.04)
Constant	-0.974***	-0.880***	-0.843***	-2.066***
	(0.01)	(0.01)	(0.01)	(0.03)
Controls	No	Yes	Yes	Yes
Observations	663,951	640,246	640,246	545,426
AIC	651,014	619,072	617,811	504,875

Notes: Schurer et al., *Integrated Census Microdata (I-CeM), 1851-1911*; Schurer and Higgs, *Integrated Census Microdata (I-CeM) Names and Addresses, 1851-1911*; Marco H.D. Van Leeuwen and Maas, "Historical Studies of Social Mobility and Stratification"; Lambert et al., "The Construction of HISCAM," pp. 77-89

Table 15: Definitions of Simplified HISCLASS Groupings

HISCLASS	Simplified HISCLASS
Farmers and fishermen	Farmer
Higher managers	Higher White Collar
Higher professionals	Higher White Collar
Lower managers	Lower White Collar
Lower professionals, clerical and sales personnel	Lower White Collar
Lower clerical and sales personnel	Lower White Collar
Medium-skilled workers	Skilled
Foremen	Semiskilled
Lower-skilled workers	Semiskilled
Lower-skilled farm workers	Semiskilled
Unskilled workers	Unskilled
Unskilled farm workers	Unskilled

Notes: Schurer et al., *Integrated Census Microdata (I-CeM), 1851-1911*; Schurer and Higgs, *Integrated Census Microdata (I-CeM) Names and Addresses, 1851-1911*; Marco HD Van Leeuwen and Maas, *HISCLASS: A historical international social class scheme*.

Table 16: Definitions of Comparable Occupations Used for HISCAM Measure

HISCO	HISCO Description	OCCODE	OCCODE Description	Comparable Occupation	HISCAM-GB (Comparable Occupation)
2200	Civil engineers	68	civil engineers assistants	Civil engineers	77.44
4260	Ships officers and pilots – Ship and boat pilots	161	pilots of ships	Ships officers and pilots – Other ships,Äô officers and pilot	50.29
7220	Nurses nec – Medical nurses, untrained or level of training unknown	47	hospital sick nurses	Medical, dental, veterinary and related workers not elsewhere classified – Other medical, dental, veterinary or related worker	55.18
7320	Midwives	45	midwives	Medical, dental, veterinary and related workers not elsewhere classified – Other medical, dental, veterinary or related worker	55.18
12120	Lawyers ,Äì Barristers	38	barristers (private) advocate	Teacher (higher education)	71.37
12130	Lawyers ,Äì Solicitors	39	solicitors (private)	Teacher (higher education)	71.37
13920	Teachers nec – Governess	83	governesses (domestic)(resident)	Teachers not in higher education	65.73
13920	Teachers nec – Governess	107	governesses (domestic)(non-resident)	Teachers not in higher education	65.73
14120	Ministers of religion and members of religious orders – Ministers of religion	31	clergymen (episcopalian)	Ministers of religion and members of religious orders – Ministers of religion	74.23
15100	Authors and writers nec	56	authors editors journalists and creative advertising workers	Visual artists and art teachers – Sculptors	59.83
15230	Editors, journalists, reporters and correspondents – Journalists, Reporters, and Correspondents	57	reporters and shorthand writers	Visual artists and art teachers – Sculptors	59.83
16110	Visual artists and art teachers – Artist nfs or art teacher	79	art, music, theatre, cinema ,Äì service etc	Visual artists and art teachers – Sculptors	59.83
16110	Visual artists and art teachers – Artist nfs or art teacher	783	art students		
16400	Engravers	74	map chart geographical engravers	Visual artists and art teachers – Engraver or etcher (artistic)	64.83
21290	Contractors and builders – Other specified contractors	429	railway contractors	Contractors and builders – Construction contractors	58.7
21290	Contractors and builders – Other specified contractors	434	road contractors	Contractors and builders – Construction contractors	58.7
24100	Inspectors	125	inspectors and supervisors [officials, clerks]	Contractors and builders – Construction contractors	58.7
33210	Railway clerks and railway agents – Railway clerks	126	ticket collectors	Transport conductors – Bus, tram or streetcar conductor	54.72
38010	Telephone and telegraph operators	1	telegraph, telephone service (post office)	Postal and related clerks, mail carriers, and messengers – Postal, mail, or telegraph clerk	63.13
38010	Telephone and telegraph operators	172	telegraph, telephone service (not government)	Postal and related clerks, mail carriers, and messengers – Postal, mail, or telegraph clerk	63.13
43010	Agent, brokers, and commercial travellers – Agent, nfs	112	brokers, factors, commercial agents (not - mine, quarry, insurance)	Salespeople ,Äì Sales people, wholesale or retail trade	64.93

43030	Manufacturer,Äôs and sales agents – Agents, brokers and commercial travellers	547	newspaper agents	Salespeople ,Äì Sales people, wholesale or retail trade	64.93
43090	Agent, brokers, and commercial travellers – Other specialized agents	113	salesmen and buyers	Salespeople ,Äì Sales people, wholesale or retail trade	64.93
43090	Agent, brokers, and commercial travellers – Other specialized agents	114	commercial travellers and manufacturers' agents	Salespeople ,Äì Sales people, wholesale or retail trade	64.93
44110	Insurance, Real Estate and Securities Salesmen – Insurance, Real Estate or Securities Salesmen, nfs	122	brokers and agents finance bill discounters	Insurance, Real Estate and Securities Salesmen – Insurance salesmen and agents	63.2
49040	Street Vendors, Canvassers and News Vendors – Newsvendors	763	news boys vendors	Street Vendors, Canvassers and News Vendors – Street sellers, pedlars and hawkers	46.09
52030	Housekeepers and housekeeping service supervisors – Steward	159	merchant service - seamen (ship stewards, cooks and others)	Servants, maids and housekeeping service workers nec – Servants nfs	48.43
53101	Cooks – Cook (domestic)	101	cooks (domestic: non-resident)	Servants, maids and housekeeping service workers nec – Servants nfs	48.43
53102	Cooks – Cook (not domestic)	98	cooks (not domestic)	Servants, maids and housekeeping service workers nec – Servants nfs	48.43
53102	Cooks – Cook (not domestic)	99	cooks (college: non-resident)	Servants, maids and housekeeping service workers nec – Servants nfs	48.43
53102	Cooks – Cook (not domestic)	100	cooks (boarding lodging house: non-resident)	Servants, maids and housekeeping service workers nec – Servants nfs	48.43
53230	Waiters, bartenders and related workers – Bartender	717	barmen (not in service)	Waiters, bartenders and related workers – Other food and drink service workers	50.41
55100	Caretakers and janitors	96	hospital servants' registry office keepers	Caretakers and janitors	58.56
55250	Charworkers, cleaners and related workers – Refuse collectors and removers	728	scavenging, street cleaners, crossing sweepers, dustmen	Charworkers, cleaners and related workers – Charworker	44.11
57020	Hairdresser, barbers, beauticians and related workers – Hairdressers	668	hairdressers	Other service workers – Service workers nec	55.22
57040	Hairdresser, barbers, beauticians and related workers – Bath attendants	106	bathing service	Other service workers – Service workers nec	55.22
58500	Prison keepers and guards	7	prison officers	Other protective service workers – Protective service workers nec	57.52
58500	Prison keepers and guards	91	prison officers (reformatory school)	Other protective service workers – Protective service workers nec	57.52
58500	Prison keepers and guards	92	prison service messengers etc (reformatory school)	Other protective service workers – Protective service workers nec	57.52
61400	Farm Managers – Farm managers, foremen and supervisors	177	farm - bailiffs stewards foremen	Specialized farmers – Livestock farmers	56.59
62113	Farm labourers and helpers, general farming and nfs – Farmer's sons and other relatives	175	farmers sons grandsons nephews	Farm labourers: livestock workers and non-farm livestock caretakers – Horse workers	49.45
62710	Farm labourers: garden and nursery workers, including non-farm gardeners – Gardeners (non-domestic)	185	other gardeners (not domestic)	Farm labourers: livestock workers and non-farm livestock caretakers – Cattle workers, except specified dairy	46.27
62711	Farm labourers: garden and nursery workers, including non-farm gardeners – domestic gardeners	87	domestic gardeners	Textile workers, specialization unknown	45.72

64970	Other related workers in fishing and hunting nec – Trapper or hunter	189	rabbit catchers trappers destroyers (on farm)	Other related workers in fishing and hunting nec – Other fishermen, hunters and related workers	41.08
64970	Other related workers in fishing and hunting nec – Trapper or hunter	191	vermin destroyers (agriculture)	Other related workers in fishing and hunting nec – Other fishermen, hunters and related workers	41.08
73300	Papermill machine operators and paper makers	519	paper manufacture	Papermill machine operators and paper makers – Other paper product makers	57.22
73300	Papermill machine operators and paper makers	520	paper stainers	Papermill machine operators and paper makers – Other paper product makers	57.22
74320	Charcoal and coal product makers – Charcoal burners and makers	749	charcoal burners	Chemical manufacturing workers nec – Other specified chemical workers	48
74390	Charcoal and coal product makers – Other coal product makers	193	others in coal coke peat and charcoal (inc. agricultural: peat/turf workers)	Chemical manufacturing workers nec – Other specified chemical workers	48
74390	Charcoal and coal product makers – Other coal product makers	204	coke burners	Chemical manufacturing workers nec – Other specified chemical workers	48
74390	Charcoal and coal product makers – Other coal product makers	205	factory labourers (undefined) coke and gas	Chemical manufacturing workers nec – Other specified chemical workers	48
74390	Charcoal and coal product makers – Other coal product makers	206	patent fuel manufacture	Chemical manufacturing workers nec – Other specified chemical workers	48
74430	Other mineral-based product makers – Alkali and soda makers	481	alkali manufacture	Chemical manufacturing workers nec – Other specified chemical workers	48
74620	Paint and dye manufacturing workers – Paint and varnish makers	496	varnish maker	Chemical manufacturing workers nec – Fertilizer or manure maker	48
74620	Paint and dye manufacturing workers – Paint and varnish makers	498	oil and colourmen	Chemical manufacturing workers nec – Fertilizer or manure maker	48
74640	Paint and dye manufacturing workers – Ink, blacking, coloring etc makers	474	dye and paint manufacture	Chemical manufacturing workers nec – Fertilizer or manure maker	48
74640	Paint and dye manufacturing workers – Ink, blacking, coloring etc makers	475	ink and blacking manufacture	Chemical manufacturing workers nec – Fertilizer or manure maker	48
74720	Animal fat and bone produce makers – Tallow chandlers, candle makers and grease	486	tallow chandlers candle and grease manufacture	Chemical manufacturing workers nec – Other specified chemical workers	48
74730	Animal fat and bone produce makers – Soap or perfume maker	487	soap boilers and makers	Chemical manufacturing workers nec – Other specified chemical workers	48
74740	Animal fat and bone produce makers – Glue, size, gelatine makers	491	glue size and gelatine manufacture	Chemical manufacturing workers nec – Other specified chemical workers	48
75200	Spinners and winders – Spinners, doublers, twisters and winders	549	cotton & cotton goods manufacture spinning processes	Textile workers, specialization unknown	45.72
75200	Spinners and winders – Spinners, doublers, twisters and winders	550	cotton & cotton goods manufacture winding warping processes	Textile workers, specialization unknown	45.72
75200	Spinners and winders – Spinners, doublers, twisters and winders	559	wool spinners wool piecers	Textile workers, specialization unknown	45.72
75200	Spinners and winders – Spinners, doublers, twisters and winders	560	worsted and stuff manufacture spinners piecers	Textile workers, specialization unknown	45.72
75200	Spinners and winders – Spinners, doublers, twisters and winders	561	wool winders wool warpers wool weavers	Textile workers, specialization unknown	45.72

75200	Spinners and winders – Spinners, doublers, twisters and winders	562	worsted and stuff manufacture winders warpers weavers	Textile workers, specialization unknown	45.72
75200	Spinners and winders – Spinners, doublers, twisters and winders	574	silk workers - spinners	Textile workers, specialization unknown	45.72
75920	Specialized textile workers nec – Net makers	586	net manufacture (various)	Specialized textile workers nec – Other specialised textile workers	56
76140	Leather manufacturing workers – Leather curriers and finishers	506	tanners fellmongers	Leather goods manufacturing workers – Other leather goods makers	50.31
76140	Leather manufacturing workers – Leather curriers and finishers	507	curriers	Leather goods manufacturing workers – Other leather goods makers	50.31
77400	Cannery workers and other food preservers	678	provision curers	Food and beverage processors nec – Other food and beverage processors	50.99
77400	Cannery workers and other food preservers	683	fish curers	Food and beverage processors nec – Other food and beverage processors	50.99
77400	Cannery workers and other food preservers	694	jam preserve sweet makers	Food and beverage processors nec – Other food and beverage processors	50.99
77400	Cannery workers and other food preservers	700	mustard vinegar spice pickle makers	Food and beverage processors nec – Other food and beverage processors	50.99
79330	Hat manufacturing workers – Hat maker	647	felt hat manufacture wollen bonnet manufacture	Hat manufacturing workers – Other hat maker	53.09
79330	Hat manufacturing workers – Hat maker	648	cloth hat cap manufacture	Hat manufacturing workers – Other hat maker	53.09
79330	Hat manufacturing workers – Hat maker	649	hat cap (not cloth felt straw) manufacture	Hat manufacturing workers – Other hat maker	53.09
79330	Hat manufacturing workers – Hat maker	651	hatters	Hat manufacturing workers – Other hat maker	53.09
79340	Hat manufacturing workers – Straw hat maker	646	straw hat manufacture	Hat manufacturing workers – Milliner	50.31
79540	Seamstresses, sewing workers nfs and embroiderers – Embroiderers	554	muslin embroiderer	Seamstresses, sewing workers nfs and embroiderers – Other hand sewers	47.38
79540	Seamstresses, sewing workers nfs and embroiderers – Embroiderers	602	embroiderers	Seamstresses, sewing workers nfs and embroiderers – Other hand sewers	47.38
79940	Cloth and related product manufacturing workers nec – Artificial flower makers	667	artificial flower makers	Cloth and related product manufacturing workers nec – Other cloth and related product manufacturing workers	47.38
79990	Cloth and related product manufacturing workers nec – Other cloth and related product manufacturing workers	590	other fibrous materials makers	Cloth and related product manufacturing workers nec – Other cloth and related product manufacturing workers	47.38
81300	Coach, carriage, cart and wagon makers	360	railway carriage makers	Wooden product manufacturing workers nec – Other makers of wooden products	49.63
81300	Coach, carriage, cart and wagon makers	361	tram car makers	Wooden product manufacturing workers nec – Other makers of wooden products	49.63
81300	Coach, carriage, cart and wagon makers	365	coach makers	Wooden product manufacturing workers nec – Other makers of wooden products	49.63
81300	Coach, carriage, cart and wagon makers	367	cartwrights	Wooden product manufacturing workers nec – Other makers of wooden products	49.63
81400	Wheelwrights	156	wheelchairs	Wooden product manufacturing workers nec – Other makers of wooden products	49.63
81400	Wheelwrights	366	wheelwrights	Wooden product manufacturing workers nec – Other makers of wooden products	49.63

81500	Cooperage and related manufacturing workers – Coopers, hoop makers and benders	457	coopers hoop makers and benders	Wooden product manufacturing workers nec – Other makers of wooden products	49.63
81710	Wooden and paper box manufacturing workers – Box makers nfs	456	box and case makers	Wooden product manufacturing workers nec – Other makers of wooden products	49.63
81730	Wooden and paper box manufacturing workers – Paper box maker	526	paper box makers	Wooden product manufacturing workers nec – Other makers of wooden products	49.63
81990	Wooden product manufacturing workers nec – Other makers of wooden products	288	printing reglet makers	Wooden product manufacturing workers nec – Other makers of wooden products	49.63
83150	Blacksmiths and related workers – Farrier or horse shoer	261	farriers	Blacksmiths and related workers – Blacksmith	46.09
83160	Blacksmiths and related workers – Specialized makers of forged metal products	292	file makers	Blacksmiths and related workers – Blacksmith	46.09
83160	Blacksmiths and related workers – Specialized makers of forged metal products	307	anchor & chain manufacture	Blacksmiths and related workers – Blacksmith	46.09
83230	Tool and pattern makers and related workers – Pattern makers nfs and nec	253	patternmakers (engine and machine making (undefined - not textile)	Tool and pattern makers and related workers – Other toolmakers and metal markers	46.09
83230	Tool and pattern makers and related workers – Pattern makers nfs and nec	254	patternmakers (spinning weaving machinery making)	Tool and pattern makers and related workers – Other toolmakers and metal markers	46.09
83230	Tool and pattern makers and related workers – Pattern makers nfs and nec	255	patternmakers (agricultural machine and implement making)	Tool and pattern makers and related workers – Other toolmakers and metal markers	46.09
83230	Tool and pattern makers and related workers – Pattern makers nfs and nec	256	patternmakers (domestic machinery making)	Tool and pattern makers and related workers – Other toolmakers and metal markers	46.09
83230	Tool and pattern makers and related workers – Pattern makers nfs and nec	257	patternmakers (undefined)	Tool and pattern makers and related workers – Other toolmakers and metal markers	46.09
83600	Firearms manufacturing workers – Gunsmith	301	gunsmiths	Cutlers, Metal Grinders, Polishers and Sharpeners – Cutlers and cutting instrument makers	45.27
83700	Lock and key manufacturing workers – Locksmith	315	lock, key makers	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83700	Lock and key manufacturing workers – Locksmith	427	locksmiths bellhangers	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83820	Metal product manufacturing workers nec – Sawmakers	293	saw makers	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83830	Metal product manufacturing workers nec – Agricultural implement and related products manufacturing workers	289	toolmakers (agricultural machine and implement making)	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83850	Metal product manufacturing workers nec – Nailers and nail makers	305	nail manufacture	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83860	Metal product manufacturing workers nec – Wire makers	312	wire/spring mattress makers	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83860	Metal product manufacturing workers nec – Wire makers	313	wire makers workers weavers	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83860	Metal product manufacturing workers nec – Wire makers	314	wire fencer	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83890	Metal product manufacturing workers nec – Other metal workers nec	295	needle makers	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04

83890	Metal product manufacturing workers nec – Other metal workers nec	296	pin makers	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83890	Metal product manufacturing workers nec – Other metal workers nec	297	steel pen makers	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83890	Metal product manufacturing workers nec – Other metal workers nec	300	die, seal, coin, medal ,Äì maker	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83890	Metal product manufacturing workers nec – Other metal workers nec	306	bolt nut rivet and screw manufacture	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83890	Metal product manufacturing workers nec – Other metal workers nec	308	gas stove makers	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83890	Metal product manufacturing workers nec – Other metal workers nec	309	stove grate range fire-iron makers	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83890	Metal product manufacturing workers nec – Other metal workers nec	310	brass bedstead makers	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83890	Metal product manufacturing workers nec – Other metal workers nec	311	iron bedstead makers	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83890	Metal product manufacturing workers nec – Other metal workers nec	316	gas fittings makers	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83890	Metal product manufacturing workers nec – Other metal workers nec	317	lamp lantern candlestick makers	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83890	Metal product manufacturing workers nec – Other metal workers nec	319	pewter white metal plated ware manufacture	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83890	Metal product manufacturing workers nec – Other metal workers nec	323	leaden goods manufacture (various)	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83890	Metal product manufacturing workers nec – Other metal workers nec	324	zinc goods workers	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83890	Metal product manufacturing workers nec – Other metal workers nec	325	brass bronze implement makers	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83890	Metal product manufacturing workers nec – Other metal workers nec	326	brass, bronze goods workers	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83890	Metal product manufacturing workers nec – Other metal workers nec	327	brass clasp buckle hinge makers	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83890	Metal product manufacturing workers nec – Other metal workers nec	328	iron domestic implement makers	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83890	Metal product manufacturing workers nec – Other metal workers nec	329	iron clasp buckle hinge makers	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83890	Metal product manufacturing workers nec – Other metal workers nec	330	iron fence and gate maker	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83890	Metal product manufacturing workers nec – Other metal workers nec	331	other iron goods makers	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83890	Metal product manufacturing workers nec – Other metal workers nec	332	fire proof safe maker	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83890	Metal product manufacturing workers nec – Other metal workers nec	333	spring maker	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04

83890	Metal product manufacturing workers nec – Other metal workers nec	335	other implement makers	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83890	Metal product manufacturing workers nec – Other metal workers nec	338	fancy chain ring gilt toy makers	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83890	Metal product manufacturing workers nec – Other metal workers nec	339	other metal workers	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83890	Metal product manufacturing workers nec – Other metal workers nec	340	mixed or unspecified metals - tube manufacture	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83890	Metal product manufacturing workers nec – Other metal workers nec	342	clasp buckle hinge makers - not brass or iron	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83890	Metal product manufacturing workers nec – Other metal workers nec	344	shackle makers	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83890	Metal product manufacturing workers nec – Other metal workers nec	411	stove setters furnace oven liners	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
83890	Metal product manufacturing workers nec – Other metal workers nec	442	house & shop fittings makers	Sheet metal manufacturing and related workers – Other sheet metal workers	53.04
84120	Mechanics, millwrights and machine manufacturing workers – Millwright	258	millwrights	Mechanics, millwrights and machine manufacturing workers – Other machinery fitters and machine assemblers	49.5
84130	Mechanics, millwrights and machine manufacturing workers – Machine makers, builders and fitters	263	fitters, turners (engine and machine)	Mechanics, millwrights and machine manufacturing workers – Other machinery fitters and machine assemblers	49.5
84130	Mechanics, millwrights and machine manufacturing workers – Machine makers, builders and fitters	264	colliery fitters	Mechanics, millwrights and machine manufacturing workers – Other machinery fitters and machine assemblers	49.5
84130	Mechanics, millwrights and machine manufacturing workers – Machine makers, builders and fitters	265	railway - signal switch turntable fitters	Mechanics, millwrights and machine manufacturing workers – Other machinery fitters and machine assemblers	49.5
84130	Mechanics, millwrights and machine manufacturing workers – Machine makers, builders and fitters	266	fitters, turners (engine and machine) labourers	Mechanics, millwrights and machine manufacturing workers – Other machinery fitters and machine assemblers	49.5
84130	Mechanics, millwrights and machine manufacturing workers – Machine makers, builders and fitters	443	refrigerator maker	Mechanics, millwrights and machine manufacturing workers – Other machinery fitters and machine assemblers	49.5
84290	Watch, clock and precision instrument manufacturing workers – Other specialized instrument makers (not musical instruments)	387	philosophical instrument makers (scientific and optical)	Watch, clock and precision instrument manufacturing workers – Watch and clock makers	55.13
84290	Watch, clock and precision instrument manufacturing workers – Other specialized instrument makers (not musical instruments)	388	photographic apparatus makers	Watch, clock and precision instrument manufacturing workers – Watch and clock makers	55.13
84290	Watch, clock and precision instrument manufacturing workers – Other specialized instrument makers (not musical instruments)	389	weighing and measuring machine makers	Watch, clock and precision instrument manufacturing workers – Watch and clock makers	55.13

84290	Watch, clock and precision instrument manufacturing workers – Other specialized instrument makers (not musical instruments)	390	tinmen (meter making)	Watch, clock and precision instrument manufacturing workers – Watch and clock makers	55.13
84290	Watch, clock and precision instrument manufacturing workers – Other specialized instrument makers (not musical instruments)	391	dental instrument and apparatus makers	Watch, clock and precision instrument manufacturing workers – Watch and clock makers	55.13
84290	Watch, clock and precision instrument manufacturing workers – Other specialized instrument makers (not musical instruments)	392	surgical instrument and apparatus makers	Watch, clock and precision instrument manufacturing workers – Watch and clock makers	55.13
84460	Motor vehicle manufacturing and repair workers – Automobile manufacturing workers	364	motor car body makers	Motor vehicle manufacturing and repair workers – Motor vehicle manufacturing and repair workers, specialization unknown	54.27
87520	Ship and boat construction workers – Ship and boat builders	350	wood ships - worker in wood	Building construction workers nec – Construction workers nec	59.38
87530	Ship and boat construction workers – Shipwright or ship joiner	345	ship boat platers rivetters	Building construction workers nec – Construction workers nec	59.38
87530	Ship and boat construction workers – Shipwright or ship joiner	346	metal ships - workers in iron	Building construction workers nec – Construction workers nec	59.38
87530	Ship and boat construction workers – Shipwright or ship joiner	348	shipwrights - wood ships	Building construction workers nec – Construction workers nec	59.38
87530	Ship and boat construction workers – Shipwright or ship joiner	349	shipwrights - metal ships	Building construction workers nec – Construction workers nec	59.38
87550	Ship and boat construction workers – Block, mast, and tackle maker	351	mast, yard, oar, block maker	Building construction workers nec – Construction workers nec	59.38
87590	Ship and boat construction workers – Other ship builders	347	fitters (ships)	Building construction workers nec – Construction workers nec	59.38
87590	Ship and boat construction workers – Other ship builders	352	ship boat painters (wood)	Building construction workers nec – Construction workers nec	59.38
87590	Ship and boat construction workers – Other ship builders	353	ship boat painters (iron)	Building construction workers nec – Construction workers nec	59.38
87590	Ship and boat construction workers – Other ship builders	354	shipyard labourers (undefined)	Building construction workers nec – Construction workers nec	59.38
87590	Ship and boat construction workers – Other ship builders	355	others in ship/boat building - wood (default)	Building construction workers nec – Construction workers nec	59.38
87590	Ship and boat construction workers – Other ship builders	356	others in ship/boat building - metal	Building construction workers nec – Construction workers nec	59.38
88020	Jewellery and precious metal workers – Jeweller	382	gold and silversmiths jewellers (not dealers)	Jewellery and precious metal workers – Silversmiths	65.43
89130	Glass manufacturing workers – Glass bottle makers	469	glass bottle manufacture	Glass manufacturing workers – Glass makers	33.04
89620	Lime, plaster and cement manufacturing workers – Cement makers	465	plaster and cement manufacture	Brick and tile manufacturing workers	37.4
89630	Lime, plaster and cement manufacturing workers – Lime burners	229	lime burners	Brick and tile manufacturing workers	37.4
91020	Paper and paperboard products makers – Paper and paperboard products makers except boxes	522	card, stationery makers	Other production and related workers – Other production and related workers nec	49.68

91020	Paper and paperboard products makers – Paper and paperboard products makers except boxes	523	valentine maker	Other production and related workers – Other production and related workers nec	49.68
91020	Paper and paperboard products makers – Paper and paperboard products makers except boxes	524	envelope manufacture	Other production and related workers – Other production and related workers nec	49.68
91020	Paper and paperboard products makers – Paper and paperboard products makers except boxes	525	paper bag makers	Other production and related workers – Other production and related workers nec	49.68
92800	Textile printing workers – Textile printers	614	wool woollen goods printers	Printing workers nec – Other printers and related workers	55.75
92800	Textile printing workers – Textile printers	615	silk printers	Printing workers nec – Other printers and related workers	55.75
92800	Textile printing workers – Textile printers	616	cotton & calico printers	Printing workers nec – Other printers and related workers	55.75
92800	Textile printing workers – Textile printers	617	undefined textile printers	Printing workers nec – Other printers and related workers	55.75
93330	Lacquerers, enamellers and japanners	742	japanners	Painting workers – Other specialized painters nec	57.11
93400	Gilders	446	gilders	Painting workers – Other specialized painters nec	57.11
94300	Non-metallic mineral product manufacturing workers – Non-metallic mineral product maker	234	asbestos maker	Other production and related workers – Other production and related workers nec	44.54
95110	Bricklayers, stonemasons and tile setters – Mason nfs	414	masons	Bricklayers, stonemasons and tile setters – Stone masons	50
95110	Bricklayers, stonemasons and tile setters – Mason nfs	415	masons labourers	Bricklayers, stonemasons and tile setters – Stone masons	50
95330	Roofers ,Äì Thatchers	192	thatchers (agriculture)	Roofers – Slate and tile roofers	42.84
95330	Roofers ,Äì Thatchers	406	thatchers (not agriculture)	Roofers – Slate and tile roofers	42.84
97220	Riggers and cable splicers – Ship riggers	357	riggers (ships)	Road, railway, waterway and related construction labourers (excavators and earth moving road workers) ,Äì Waterway/harbour builders, workers & lab	42.24
97410	Road, railway, waterway and related construction labourers (excavators and earth moving road workers) – Navvy, excavator and digger nfs	408	builders' excavators	Road, railway, waterway and related construction labourers (excavators and earth moving road workers) ,Äì Waterway/harbour builders, workers & lab	42.24
97430	Road, railway, waterway and related construction labourers (excavators and earth moving road workers) – Railway builders, workers and labourers	131	platelayers gangers packers	Road, railway, waterway and related construction labourers (excavators and earth moving road workers) ,Äì Waterway/harbour builders, workers & lab	42.24
97430	Road, railway, waterway and related construction labourers (excavators and earth moving road workers) – Railway builders, workers and labourers	199	railway labourers navvies (coal mine)	Road, railway, waterway and related construction labourers (excavators and earth moving road workers) ,Äì Waterway/harbour builders, workers & lab	42.24
97490	Road, railway, waterway and related construction labourers (excavators and earth moving road workers) – Other	188	land drainage service	Road, railway, waterway and related construction labourers (excavators and earth moving road workers) ,Äì Waterway/harbour builders, workers & lab	42.24
97490	Road, railway, waterway and related construction labourers (excavators and earth moving road workers) – Other	430	pond reservoir makers	Road, railway, waterway and related construction labourers (excavators and earth moving road workers) ,Äì Waterway/harbour builders, workers & lab	42.24
97490	Road, railway, waterway and related construction labourers (excavators and earth moving road workers) – Other	432	railway labourers navvies (contractors labourers) default	Road, railway, waterway and related construction labourers (excavators and earth moving road workers) ,Äì Waterway/harbour builders, workers & lab	42.24

97490	Road, railway, waterway and related construction labourers (excavators and earth moving road workers) – Other	436	road labourers	Road, railway, waterway and related construction labourers (excavators and earth moving road workers) ,Äì Waterway/harbour builders, workers & lab	42.24
97490	Road, railway, waterway and related construction labourers (excavators and earth moving road workers) – Other	727	town drainage	Road, railway, waterway and related construction labourers (excavators and earth moving road workers) ,Äì Waterway/harbour builders, workers & lab	42.24
98200	Ship's engine men	158	merchant service - seamen (engineering)	Sailors and boatmen – Seamen	41.75
98490	Other skilled railway workers – Other railway workers	132	railway labourer (not contractor's labourer)	Other skilled railway workers – Railway signallers	47.79
98490	Other skilled railway workers – Other railway workers	134	miscellaneous servants on railway	Other skilled railway workers – Railway signallers	47.79
98900	Other transport equipment operators	151	omnibus (not drivers conductors)	Railway/Railroad Locomotive operators	42.15
98900	Other transport equipment operators	154	tramway service not drivers conductors	Railway/Railroad Locomotive operators	42.15
99120	Working nfs – Labourers nfs	764	corporation borough council labourers (undefined)	Other production and related workers – Other production and related workers nec	44.54
99130	Working, nfs – Common labourers or general labourers	765	general labourers	Other production and related workers – Other production and related workers nec	44.54
99150	Working, nfs	769	apprentices	Other production and related workers – Other production and related workers nec	44.54
99200	Factory labourers, unspecified	770	factory labourers (undefined)	Other production and related workers – Other production and related workers nec	44.54